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Series Solutions of Linear Differential Equations

EXERCISES 5.1

Solutions About Ordinary Points

$$1. \lim_{n \rightarrow \infty} \left| \frac{2^{n+1}x^{n+1}/(n+1)}{2^n x^n/n} \right| = \lim_{n \rightarrow \infty} \frac{2n}{n+1} |x| = 2|x|$$

The series is absolutely convergent for $2|x| < 1$ or $|x| < \frac{1}{2}$. The radius of convergence is $R = \frac{1}{2}$. At $x = -\frac{1}{2}$, the series $\sum_{n=1}^{\infty} (-1)^n/n$ converges by the alternating series test. At $x = \frac{1}{2}$, the series $\sum_{n=1}^{\infty} 1/n$ is the harmonic series which diverges. Thus, the given series converges on $[-\frac{1}{2}, \frac{1}{2})$.

$$2. \lim_{n \rightarrow \infty} \left| \frac{100^{n+1}(x+7)^{n+1}/(n+1)!}{100^n(x+7)^n/n!} \right| = \lim_{n \rightarrow \infty} \frac{100}{n+1} |x+7| = 0$$

The radius of convergence is $R = \infty$. The series is absolutely convergent on $(-\infty, \infty)$.

3. By the ratio test,

$$\lim_{n \rightarrow \infty} \left| \frac{(x-5)^{n+1}/10^{n+1}}{(x-5)^n/10^n} \right| = \lim_{n \rightarrow \infty} \frac{1}{10} |x-5| = \frac{1}{10} |x-5|.$$

The series is absolutely convergent for $\frac{1}{10}|x-5| < 1$, $|x-5| < 10$, or on $(-5, 15)$. The radius of convergence is $R = 10$. At $x = -5$, the series $\sum_{n=1}^{\infty} (-1)^n(-10)^n/10^n = \sum_{n=1}^{\infty} 1$ diverges by the n th term test. At $x = 15$, the series $\sum_{n=1}^{\infty} (-1)^n 10^n/10^n = \sum_{n=1}^{\infty} (-1)^n$ diverges by the n th term test. Thus, the series converges on $(-5, 15)$.

$$4. \lim_{n \rightarrow \infty} \left| \frac{(n+1)!(x-1)^{n+1}}{n!(x-1)^n} \right| = \lim_{n \rightarrow \infty} (n+1)|x-1| = \begin{cases} \infty, & x \neq 1 \\ 0, & x = 1 \end{cases}$$

The radius of convergence is $R = 0$ and the series converges only for $x = 1$.

$$5. \sin x \cos x = \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \frac{x^7}{5040} + \cdots \right) \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots \right) = x - \frac{2x^3}{3} + \frac{2x^5}{15} - \frac{4x^7}{315} + \cdots$$

$$6. e^{-x} \cos x = \left(1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} - \cdots \right) \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots \right) = 1 - x + \frac{x^3}{3} - \frac{x^4}{6} + \cdots$$

$$7. \frac{1}{\cos x} = \frac{1}{1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} + \cdots} = 1 + \frac{x^2}{2} + \frac{5x^4}{24} + \frac{61x^6}{720} + \cdots$$

Since $\cos(\pi/2) = \cos(-\pi/2) = 0$, the series converges on $(-\pi/2, \pi/2)$.

$$8. \frac{1-x}{2+x} = \frac{1}{2} - \frac{3}{4}x + \frac{3}{8}x^2 - \frac{3}{16}x^3 + \cdots$$

Since the function is undefined at $x = -2$, the series converges on $(-2, 2)$.

9. Let $k = n + 2$ so that $n = k - 2$ and

$$\sum_{n=1}^{\infty} n c_n x^{n+2} = \sum_{k=3}^{\infty} (k-2) c_{k-2} x^k.$$

10. Let $k = n - 3$ so that $n = k + 3$ and

$$\sum_{n=3}^{\infty} (2n-1)c_n x^{n-3} = \sum_{k=0}^{\infty} (2k+5)c_{k+3} x^k.$$

$$\begin{aligned} 11. \quad \sum_{n=1}^{\infty} 2nc_n x^{n-1} + \sum_{n=0}^{\infty} 6c_n x^{n+1} &= 2 \cdot 1 \cdot c_1 x^0 + \underbrace{\sum_{n=2}^{\infty} 2nc_n x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=0}^{\infty} 6c_n x^{n+1}}_{k=n+1} \\ &= 2c_1 + \sum_{k=1}^{\infty} 2(k+1)c_{k+1} x^k + \sum_{k=1}^{\infty} 6c_{k-1} x^k \\ &= 2c_1 + \sum_{k=1}^{\infty} [2(k+1)c_{k+1} + 6c_{k-1}] x^k \end{aligned}$$

$$\begin{aligned} 12. \quad \sum_{n=2}^{\infty} n(n-1)c_n x^n + 2 \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + 3 \sum_{n=1}^{\infty} nc_n x^n \\ &= 2 \cdot 2 \cdot 1c_2 x^0 + 2 \cdot 3 \cdot 2c_3 x^1 + 3 \cdot 1 \cdot c_1 x^1 + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} + 2 \underbrace{\sum_{n=4}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + 3 \underbrace{\sum_{n=2}^{\infty} nc_n x^n}_{k=n} \\ &= 4c_2 + (3c_1 + 12c_3)x + \sum_{k=2}^{\infty} k(k-1)c_k x^k + 2 \sum_{k=2}^{\infty} (k+2)(k+1)c_{k+2} x^k + 3 \sum_{k=2}^{\infty} kc_k x^k \\ &= 4c_2 + (3c_1 + 12c_3)x + \sum_{k=2}^{\infty} [(k(k-1) + 3k)c_k + 2(k+2)(k+1)c_{k+2}] x^k \\ &= 4c_2 + (3c_1 + 12c_3)x + \sum_{k=2}^{\infty} [k(k+2)c_k + 2(k+1)(k+2)c_{k+2}] x^k \end{aligned}$$

$$\begin{aligned} 13. \quad y' &= \sum_{n=1}^{\infty} (-1)^{n+1} x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-2} \\ (x+1)y'' + y' &= (x+1) \sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-2} + \sum_{n=1}^{\infty} (-1)^{n+1} x^{n-1} \\ &= \sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-1} + \sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-2} + \sum_{n=1}^{\infty} (-1)^{n+1} x^{n-1} \\ &= -x^0 + x^0 + \underbrace{\sum_{n=2}^{\infty} (-1)^{n+1} (n-1) x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=3}^{\infty} (-1)^{n+1} (n-1) x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=2}^{\infty} (-1)^{n+1} x^{n-1}}_{k=n-1} \\ &= \sum_{k=1}^{\infty} (-1)^{k+2} k x^k + \sum_{k=1}^{\infty} (-1)^{k+3} (k+1) x^k + \sum_{k=1}^{\infty} (-1)^{k+2} x^k \\ &= \sum_{k=1}^{\infty} [(-1)^{k+2} k - (-1)^{k+2} k - (-1)^{k+2} + (-1)^{k+2}] x^k = 0 \end{aligned}$$

$$14. \quad y' = \sum_{n=1}^{\infty} \frac{(-1)^n 2n}{2^{2n} (n!)^2} x^{2n-1}, \quad y'' = \sum_{n=1}^{\infty} \frac{(-1)^n 2n(2n-1)}{2^{2n} (n!)^2} x^{2n-2}$$

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$$\begin{aligned}
 xy'' + y' + xy &= \underbrace{\sum_{n=1}^{\infty} \frac{(-1)^n 2n(2n-1)}{2^{2n}(n!)^2} x^{2n-1}}_{k=n} + \underbrace{\sum_{n=1}^{\infty} \frac{(-1)^n 2n}{2^{2n}(n!)^2} x^{2n-1}}_{k=n} + \underbrace{\sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n}(n!)^2} x^{2n+1}}_{k=n+1} \\
 &= \sum_{k=1}^{\infty} \left[\frac{(-1)^k 2k(2k-1)}{2^{2k}(k!)^2} + \frac{(-1)^k 2k}{2^{2k}(k!)^2} + \frac{(-1)^{k-1}}{2^{2k-2}[(k-1)!]^2} \right] x^{2k-1} \\
 &= \sum_{k=1}^{\infty} \left[\frac{(-1)^k (2k)^2}{2^{2k}(k!)^2} - \frac{(-1)^k}{2^{2k-2}[(k-1)!]^2} \right] x^{2k-1} \\
 &= \sum_{k=1}^{\infty} (-1)^k \left[\frac{(2k)^2 - 2^2 k^2}{2^{2k}(k!)^2} \right] x^{2k-1} = 0
 \end{aligned}$$

15. The singular points of $(x^2 - 25)y'' + 2xy' + y = 0$ are -5 and 5 . The distance from 0 to either of these points is 5 . The distance from 1 to the closest of these points is 4 .
16. The singular points of $(x^2 - 2x + 10)y'' + xy' - 4y = 0$ are $1 + 3i$ and $1 - 3i$. The distance from 0 to either of these points is $\sqrt{10}$. The distance from 1 to either of these points is 3 .
17. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 y'' - xy &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=0}^{\infty} c_n x^{n+1}}_{k=n+1} = \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} c_{k-1} x^k \\
 &= 2c_2 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - c_{k-1}] x^k = 0.
 \end{aligned}$$

Thus

$$c_2 = 0$$

$$(k+2)(k+1)c_{k+2} - c_{k-1} = 0$$

and

$$c_{k+2} = \frac{1}{(k+2)(k+1)} c_{k-1}, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}
 c_3 &= \frac{1}{6} \\
 c_4 &= c_5 = 0 \\
 c_6 &= \frac{1}{180}
 \end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}
 c_3 &= 0 \\
 c_4 &= \frac{1}{12} \\
 c_5 &= c_6 = 0 \\
 c_7 &= \frac{1}{504}
 \end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{6}x^3 + \frac{1}{180}x^6 + \dots \quad \text{and} \quad y_2 = x + \frac{1}{12}x^4 + \frac{1}{504}x^7 + \dots$$

18. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' + x^2 y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=0}^{\infty} c_n x^{n+2}}_{k=n+2} = \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=2}^{\infty} c_{k-2} x^k \\ &= 2c_2 + 6c_3 x + \sum_{k=2}^{\infty} [(k+2)(k+1)c_{k+2} + c_{k-2}] x^k = 0. \end{aligned}$$

Thus

$$c_2 = c_3 = 0$$

$$(k+2)(k+1)c_{k+2} + c_{k-2} = 0$$

and

$$c_{k+2} = -\frac{1}{(k+2)(k+1)} c_{k-2}, \quad k = 2, 3, 4, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned} c_4 &= -\frac{1}{12} \\ c_5 &= c_6 = c_7 = 0 \\ c_8 &= \frac{1}{672} \end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned} c_4 &= 0 \\ c_5 &= -\frac{1}{20} \\ c_6 &= c_7 = c_8 = 0 \\ c_9 &= \frac{1}{1440} \end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{12}x^4 + \frac{1}{672}x^8 - \dots \quad \text{and} \quad y_2 = x - \frac{1}{20}x^5 + \frac{1}{1440}x^9 - \dots$$

19. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' - 2xy' + y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - 2 \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - 2 \sum_{k=1}^{\infty} k c_k x^k + \sum_{k=0}^{\infty} c_k x^k \\ &= 2c_2 + c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (2k-1)c_k] x^k = 0. \end{aligned}$$

Thus

$$2c_2 + c_0 = 0$$

$$(k+2)(k+1)c_{k+2} - (2k-1)c_k = 0$$

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and

$$c_2 = -\frac{1}{2}c_0$$

$$c_{k+2} = \frac{2k-1}{(k+2)(k+1)} c_k, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = -\frac{1}{2}$$

$$c_3 = c_5 = c_7 = \dots = 0$$

$$c_4 = -\frac{1}{8}$$

$$c_6 = -\frac{7}{240}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = c_4 = c_6 = \dots = 0$$

$$c_3 = \frac{1}{6}$$

$$c_5 = \frac{1}{24}$$

$$c_7 = \frac{1}{112}$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{2}x^2 - \frac{1}{8}x^4 - \frac{7}{240}x^6 - \dots \quad \text{and} \quad y_2 = x + \frac{1}{6}x^3 + \frac{1}{24}x^5 + \frac{1}{112}x^7 + \dots$$

20. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' - xy' + 2y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + 2 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} k c_k x^k + 2 \sum_{k=0}^{\infty} c_k x^k \\ &= 2c_2 + 2c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (k-2)c_k] x^k = 0. \end{aligned}$$

Thus

$$2c_2 + 2c_0 = 0$$

$$(k+2)(k+1)c_{k+2} - (k-2)c_k = 0$$

and

$$c_2 = -c_0$$

$$c_{k+2} = \frac{k-2}{(k+2)(k+1)} c_k, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned} c_2 &= -1 \\ c_3 &= c_5 = c_7 = \cdots = 0 \\ c_4 &= 0 \\ c_6 &= c_8 = c_{10} = \cdots = 0. \end{aligned}$$

For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned} c_2 &= c_4 = c_6 = \cdots = 0 \\ c_3 &= -\frac{1}{6} \\ c_5 &= -\frac{1}{120} \end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 - x^2 \quad \text{and} \quad y_2 = x - \frac{1}{6}x^3 - \frac{1}{120}x^5 - \cdots.$$

21. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' + x^2 y' + xy &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^{n+1}}_{k=n+1} + \underbrace{\sum_{n=0}^{\infty} c_n x^{n+1}}_{k=n+1} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=2}^{\infty} (k-1)c_{k-1} x^k + \sum_{k=1}^{\infty} c_{k-1} x^k \\ &= 2c_2 + (6c_3 + c_0)x + \sum_{k=2}^{\infty} [(k+2)(k+1)c_{k+2} + k c_{k-1}] x^k = 0. \end{aligned}$$

Thus

$$\begin{aligned} c_2 &= 0 \\ 6c_3 + c_0 &= 0 \\ (k+2)(k+1)c_{k+2} + k c_{k-1} &= 0 \end{aligned}$$

and

$$\begin{aligned} c_2 &= 0 \\ c_3 &= -\frac{1}{6}c_0 \\ c_{k+2} &= -\frac{k}{(k+2)(k+1)} c_{k-1}, \quad k = 2, 3, 4, \dots \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned} c_3 &= -\frac{1}{6} \\ c_4 &= c_5 = 0 \\ c_6 &= \frac{1}{45} \end{aligned}$$

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and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_3 &= 0 \\c_4 &= -\frac{1}{6} \\c_5 &= c_6 = 0 \\c_7 &= \frac{5}{252}\end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{6}x^3 + \frac{1}{45}x^6 - \cdots \quad \text{and} \quad y_2 = x - \frac{1}{6}x^4 + \frac{5}{252}x^7 - \cdots.$$

22. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}y'' + 2xy' + 2y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + 2 \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + 2 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\&= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + 2 \sum_{k=1}^{\infty} k c_k x^k + 2 \sum_{k=0}^{\infty} c_k x^k \\&= 2c_2 + 2c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + 2(k+1)c_k] x^k = 0.\end{aligned}$$

Thus

$$\begin{aligned}2c_2 + 2c_0 &= 0 \\(k+2)(k+1)c_{k+2} + 2(k+1)c_k &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= -c_0 \\c_{k+2} &= -\frac{2}{k+2} c_k, \quad k = 1, 2, 3, \dots\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}c_2 &= -1 \\c_3 &= c_5 = c_7 = \cdots = 0 \\c_4 &= \frac{1}{2} \\c_6 &= -\frac{1}{6}\end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_2 &= c_4 = c_6 = \cdots = 0 \\c_3 &= -\frac{2}{3} \\c_5 &= \frac{4}{15} \\c_7 &= -\frac{8}{105}\end{aligned}$$

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and so on. Thus, two solutions are

$$y_1 = 1 - x^2 + \frac{1}{2}x^4 - \frac{1}{6}x^6 + \cdots \quad \text{and} \quad y_2 = x - \frac{2}{3}x^3 + \frac{4}{15}x^5 - \frac{8}{105}x^7 + \cdots.$$

23. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} (x-1)y'' + y' &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} - \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^{n-1}}_{k=n-1} \\ &= \sum_{k=1}^{\infty} (k+1)k c_{k+1} x^k - \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=0}^{\infty} (k+1)c_{k+1} x^k \\ &= -2c_2 + c_1 + \sum_{k=1}^{\infty} [(k+1)k c_{k+1} - (k+2)(k+1)c_{k+2} + (k+1)c_{k+1}] x^k = 0. \end{aligned}$$

Thus

$$-2c_2 + c_1 = 0$$

$$(k+1)^2 c_{k+1} - (k+2)(k+1)c_{k+2} = 0$$

and

$$c_2 = \frac{1}{2}c_1$$

$$c_{k+2} = \frac{k+1}{k+2} c_{k+1}, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find $c_2 = c_3 = c_4 = \cdots = 0$. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = \frac{1}{2}, \quad c_3 = \frac{1}{3}, \quad c_4 = \frac{1}{4},$$

and so on. Thus, two solutions are

$$y_1 = 1 \quad \text{and} \quad y_2 = x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \frac{1}{4}x^4 + \cdots$$

24. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} (x+2)y'' + xy' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=2}^{\infty} 2n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^{n-1}}_{k=n} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=1}^{\infty} (k+1)k c_{k+1} x^k + \sum_{k=0}^{\infty} 2(k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} k c_k x^k - \sum_{k=0}^{\infty} c_k x^k \\ &= 4c_2 - c_0 + \sum_{k=1}^{\infty} [(k+1)k c_{k+1} + 2(k+2)(k+1)c_{k+2} + (k-1)c_k] x^k = 0. \end{aligned}$$

Thus

$$4c_2 - c_0 = 0$$

$$(k+1)k c_{k+1} + 2(k+2)(k+1)c_{k+2} + (k-1)c_k = 0, \quad k = 1, 2, 3, \dots$$

and

$$c_2 = \frac{1}{4}c_0$$

$$c_{k+2} = -\frac{(k+1)k c_{k+1} + (k-1)c_k}{2(k+2)(k+1)}, \quad k = 1, 2, 3, \dots$$

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Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_1 = 0, \quad c_2 = \frac{1}{4}, \quad c_3 = -\frac{1}{24}, \quad c_4 = 0, \quad c_5 = \frac{1}{480}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned} c_2 &= 0 \\ c_3 &= 0 \\ c_4 &= c_5 = c_6 = \cdots = 0. \end{aligned}$$

Thus, two solutions are

$$y_1 = c_0 \left[1 + \frac{1}{4}x^2 - \frac{1}{24}x^3 + \frac{1}{480}x^5 + \cdots \right] \quad \text{and} \quad y_2 = c_1 x.$$

25. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' - (x+1)y' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} - \underbrace{\sum_{n=1}^{\infty} n c_n x^{n-1}}_{k=n-1} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} k c_k x^k - \sum_{k=0}^{\infty} (k+1)c_{k+1} x^k - \sum_{k=0}^{\infty} c_k x^k \\ &= 2c_2 - c_1 - c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (k+1)c_{k+1} - (k+1)c_k] x^k = 0. \end{aligned}$$

Thus

$$2c_2 - c_1 - c_0 = 0$$

$$(k+2)(k+1)c_{k+2} - (k+1)(c_{k+1} + c_k) = 0$$

and

$$\begin{aligned} c_2 &= \frac{c_1 + c_0}{2} \\ c_{k+2} &= \frac{c_{k+1} + c_k}{k+2}, \quad k = 1, 2, 3, \dots \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = \frac{1}{2}, \quad c_3 = \frac{1}{6}, \quad c_4 = \frac{1}{6},$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = \frac{1}{2}, \quad c_3 = \frac{1}{2}, \quad c_4 = \frac{1}{4},$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{6}x^4 + \cdots \quad \text{and} \quad y_2 = x + \frac{1}{2}x^2 + \frac{1}{2}x^3 + \frac{1}{4}x^4 + \cdots$$

26. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 (x^2 + 1)y'' - 6y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - 6 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\
 &= \sum_{k=2}^{\infty} k(k-1)c_k x^k + \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - 6 \sum_{k=0}^{\infty} c_k x^k \\
 &= 2c_2 - 6c_0 + (6c_3 - 6c_1)x + \sum_{k=2}^{\infty} [(k^2 - k - 6)c_k + (k+2)(k+1)c_{k+2}] x^k = 0.
 \end{aligned}$$

Thus

$$2c_2 - 6c_0 = 0$$

$$6c_3 - 6c_1 = 0$$

$$(k-3)(k+2)c_k + (k+2)(k+1)c_{k+2} = 0$$

and

$$c_2 = 3c_0$$

$$c_3 = c_1$$

$$c_{k+2} = -\frac{k-3}{k+1} c_k, \quad k = 2, 3, 4, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = 3$$

$$c_3 = c_5 = c_7 = \dots = 0$$

$$c_4 = 1$$

$$c_6 = -\frac{1}{5}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = c_4 = c_6 = \dots = 0$$

$$c_3 = 1$$

$$c_5 = c_7 = c_9 = \dots = 0.$$

Thus, two solutions are

$$y_1 = 1 + 3x^2 + x^4 - \frac{1}{5}x^6 + \dots \quad \text{and} \quad y_2 = x + x^3.$$

27. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 (x^2 + 2)y'' + 3xy' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} + 2 \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + 3 \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\
 &= \sum_{k=2}^{\infty} k(k-1)c_k x^k + 2 \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + 3 \sum_{k=1}^{\infty} k c_k x^k - \sum_{k=0}^{\infty} c_k x^k \\
 &= (4c_2 - c_0) + (12c_3 + 2c_1)x + \sum_{k=2}^{\infty} [2(k+2)(k+1)c_{k+2} + (k^2 + 2k - 1)c_k] x^k = 0.
 \end{aligned}$$

5.1 Solutions About Ordinary Points

Thus

$$\begin{aligned}4c_2 - c_0 &= 0 \\12c_3 + 2c_1 &= 0 \\2(k+2)(k+1)c_{k+2} + (k^2 + 2k - 1)c_k &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= \frac{1}{4}c_0 \\c_3 &= -\frac{1}{6}c_1 \\c_{k+2} &= -\frac{k^2 + 2k - 1}{2(k+2)(k+1)}c_k, \quad k = 2, 3, 4, \dots\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}c_2 &= \frac{1}{4} \\c_3 &= c_5 = c_7 = \dots = 0 \\c_4 &= -\frac{7}{96}\end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_2 &= c_4 = c_6 = \dots = 0 \\c_3 &= -\frac{1}{6} \\c_5 &= \frac{7}{120}\end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{4}x^2 - \frac{7}{96}x^4 + \dots \quad \text{and} \quad y_2 = x - \frac{1}{6}x^3 + \frac{7}{120}x^5 - \dots$$

28. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}(x^2 - 1)y'' + xy' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} - \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} nc_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\&= \sum_{k=2}^{\infty} k(k-1)c_k x^k - \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} kc_k x^k - \sum_{k=0}^{\infty} c_k x^k \\&= (-2c_2 - c_0) - 6c_3 x + \sum_{k=2}^{\infty} [-(k+2)(k+1)c_{k+2} + (k^2 - 1)c_k] x^k = 0.\end{aligned}$$

Thus

$$\begin{aligned}-2c_2 - c_0 &= 0 \\-6c_3 &= 0 \\-(k+2)(k+1)c_{k+2} + (k^2 - 1)c_k &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= -\frac{1}{2}c_0 \\c_3 &= 0 \\c_{k+2} &= \frac{k-1}{k+2}c_k, \quad k = 2, 3, 4, \dots\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}c_2 &= -\frac{1}{2} \\c_3 &= c_5 = c_7 = \dots = 0 \\c_4 &= -\frac{1}{8}\end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}c_2 &= c_4 = c_6 = \dots = 0 \\c_3 &= c_5 = c_7 = \dots = 0.\end{aligned}$$

Thus, two solutions are

$$y_1 = 1 - \frac{1}{2}x^2 - \frac{1}{8}x^4 - \dots \quad \text{and} \quad y_2 = x.$$

29. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}(x-1)y'' - xy' + y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} - \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\&= \sum_{k=1}^{\infty} (k+1)k c_{k+1} x^k - \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} k c_k x^k + \sum_{k=0}^{\infty} c_k x^k \\&= -2c_2 + c_0 + \sum_{k=1}^{\infty} [-(k+2)(k+1)c_{k+2} + (k+1)k c_{k+1} - (k-1)c_k] x^k = 0.\end{aligned}$$

Thus

$$\begin{aligned}-2c_2 + c_0 &= 0 \\-(k+2)(k+1)c_{k+2} + (k-1)k c_{k+1} - (k-1)c_k &= 0\end{aligned}$$

and

$$\begin{aligned}c_2 &= \frac{1}{2}c_0 \\c_{k+2} &= \frac{k c_{k+1}}{k+2} - \frac{(k-1)c_k}{(k+2)(k+1)}, \quad k = 1, 2, 3, \dots\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = \frac{1}{2}, \quad c_3 = \frac{1}{6}, \quad c_4 = 0,$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain $c_2 = c_3 = c_4 = \dots = 0$. Thus,

$$y = C_1 \left(1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots \right) + C_2 x$$

and

$$y' = C_1 \left(x + \frac{1}{2}x^2 + \dots \right) + C_2.$$

5.1 Solutions About Ordinary Points

The initial conditions imply $C_1 = -2$ and $C_2 = 6$, so

$$y = -2 \left(1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots \right) + 6x = 8x - 2e^x.$$

30. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} & (x+1)y'' - (2-x)y' + y \\ &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - 2 \underbrace{\sum_{n=1}^{\infty} n c_n x^{n-1}}_{k=n-1} + \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=1}^{\infty} (k+1)k c_{k+1} x^k + \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - 2 \sum_{k=0}^{\infty} (k+1)c_{k+1} x^k + \sum_{k=1}^{\infty} k c_k x^k + \sum_{k=0}^{\infty} c_k x^k \\ &= 2c_2 - 2c_1 + c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (k+1)c_{k+1} + (k+1)c_k] x^k = 0. \end{aligned}$$

Thus

$$2c_2 - 2c_1 + c_0 = 0$$

$$(k+2)(k+1)c_{k+2} - (k+1)c_{k+1} + (k+1)c_k = 0$$

and

$$\begin{aligned} c_2 &= c_1 - \frac{1}{2}c_0 \\ c_{k+2} &= \frac{1}{k+2} c_{k+1} - \frac{1}{k+2} c_k, \quad k = 1, 2, 3, \dots \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = -\frac{1}{2}, \quad c_3 = -\frac{1}{6}, \quad c_4 = \frac{1}{12},$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = 1, \quad c_3 = 0, \quad c_4 = -\frac{1}{4},$$

and so on. Thus,

$$y = C_1 \left(1 - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4 + \cdots \right) + C_2 \left(x + x^2 - \frac{1}{4}x^4 + \cdots \right)$$

and

$$y' = C_1 \left(-x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \cdots \right) + C_2 (1 + 2x - x^3 + \cdots).$$

The initial conditions imply $C_1 = 2$ and $C_2 = -1$, so

$$\begin{aligned} y &= 2 \left(1 - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4 + \cdots \right) - \left(x + x^2 - \frac{1}{4}x^4 + \cdots \right) \\ &= 2 - x - 2x^2 - \frac{1}{3}x^3 + \frac{5}{12}x^4 + \cdots \end{aligned}$$

31. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 y'' - 2xy' + 8y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - 2 \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + 8 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\
 &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - 2 \sum_{k=1}^{\infty} k c_k x^k + 8 \sum_{k=0}^{\infty} c_k x^k \\
 &= 2c_2 + 8c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + (8-2k)c_k] x^k = 0.
 \end{aligned}$$

Thus

$$2c_2 + 8c_0 = 0$$

$$(k+2)(k+1)c_{k+2} + (8-2k)c_k = 0$$

and

$$\begin{aligned}
 c_2 &= -4c_0 \\
 c_{k+2} &= \frac{2(k-4)}{(k+2)(k+1)} c_k, \quad k = 1, 2, 3, \dots
 \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}
 c_2 &= -4 \\
 c_3 &= c_5 = c_7 = \dots = 0 \\
 c_4 &= \frac{4}{3} \\
 c_6 &= c_8 = c_{10} = \dots = 0.
 \end{aligned}$$

For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned}
 c_2 &= c_4 = c_6 = \dots = 0 \\
 c_3 &= -1 \\
 c_5 &= \frac{1}{10}
 \end{aligned}$$

and so on. Thus,

$$y = C_1 \left(1 - 4x^2 + \frac{4}{3}x^4 \right) + C_2 \left(x - x^3 + \frac{1}{10}x^5 + \dots \right)$$

and

$$y' = C_1 \left(-8x + \frac{16}{3}x^3 \right) + C_2 \left(1 - 3x^2 + \frac{1}{2}x^4 + \dots \right).$$

The initial conditions imply $C_1 = 3$ and $C_2 = 0$, so

$$y = 3 \left(1 - 4x^2 + \frac{4}{3}x^4 \right) = 3 - 12x^2 + 4x^4.$$

5.1 Solutions About Ordinary Points

32. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 (x^2 + 1)y'' + 2xy' &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^n}_{k=n} + \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} 2nc_n x^n}_{k=n} \\
 &= \sum_{k=2}^{\infty} k(k-1)c_k x^k + \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} 2kc_k x^k \\
 &= 2c_2 + (6c_3 + 2c_1)x + \sum_{k=2}^{\infty} [k(k+1)c_k + (k+2)(k+1)c_{k+2}] x^k = 0.
 \end{aligned}$$

Thus

$$2c_2 = 0$$

$$6c_3 + 2c_1 = 0$$

$$k(k+1)c_k + (k+2)(k+1)c_{k+2} = 0$$

and

$$c_2 = 0$$

$$c_3 = -\frac{1}{3}c_1$$

$$c_{k+2} = -\frac{k}{k+2}c_k, \quad k = 2, 3, 4, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find $c_3 = c_4 = c_5 = \dots = 0$. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_3 = -\frac{1}{3}$$

$$c_4 = c_6 = c_8 = \dots = 0$$

$$c_5 = -\frac{1}{5}$$

$$c_7 = \frac{1}{7}$$

and so on. Thus

$$y = C_0 + C_1 \left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \dots \right)$$

and

$$y' = c_1 (1 - x^2 + x^4 - x^6 + \dots).$$

The initial conditions imply $c_0 = 0$ and $c_1 = 1$, so

$$y = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \frac{1}{7}x^7 + \dots$$

33. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 y'' + (\sin x)y &= \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \left(x - \frac{1}{6}x^3 + \frac{1}{120}x^5 - \dots \right) (c_0 + c_1 x + c_2 x^2 + \dots) \\
 &= [2c_2 + 6c_3 x + 12c_4 x^2 + 20c_5 x^3 + \dots] + \left[c_0 x + c_1 x^2 + \left(c_2 - \frac{1}{6}c_0 \right) x^3 + \dots \right] \\
 &= 2c_2 + (6c_3 + c_0)x + (12c_4 + c_1)x^2 + \left(20c_5 + c_2 - \frac{1}{6}c_0 \right) x^3 + \dots = 0.
 \end{aligned}$$

Thus

$$\begin{aligned} 2c_2 &= 0 \\ 6c_3 + c_0 &= 0 \\ 12c_4 + c_1 &= 0 \\ 20c_5 + c_2 - \frac{1}{6}c_0 &= 0 \end{aligned}$$

and

$$\begin{aligned} c_2 &= 0 \\ c_3 &= -\frac{1}{6}c_0 \\ c_4 &= -\frac{1}{12}c_1 \\ c_5 &= -\frac{1}{20}c_2 + \frac{1}{120}c_0. \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = 0, \quad c_3 = -\frac{1}{6}, \quad c_4 = 0, \quad c_5 = \frac{1}{120}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = 0, \quad c_3 = 0, \quad c_4 = -\frac{1}{12}, \quad c_5 = 0$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \cdots \quad \text{and} \quad y_2 = x - \frac{1}{12}x^4 + \cdots.$$

34. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' + e^x y' - y &= \sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} \\ &\quad + \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots\right) (c_1 + 2c_2x + 3c_3x^2 + 4c_4x^3 + \cdots) - \sum_{n=0}^{\infty} c_n x^n \\ &= [2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + \cdots] \\ &\quad + \left[c_1 + (2c_2 + c_1)x + \left(3c_3 + 2c_2 + \frac{1}{2}c_1\right)x^2 + \cdots\right] - [c_0 + c_1x + c_2x^2 + \cdots] \\ &= (2c_2 + c_1 - c_0) + (6c_3 + 2c_2)x + \left(12c_4 + 3c_3 + c_2 + \frac{1}{2}c_1\right)x^2 + \cdots = 0. \end{aligned}$$

Thus

$$\begin{aligned} 2c_2 + c_1 - c_0 &= 0 \\ 6c_3 + 2c_2 &= 0 \\ 12c_4 + 3c_3 + c_2 + \frac{1}{2}c_1 &= 0 \end{aligned}$$

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and

$$\begin{aligned}c_2 &= \frac{1}{2}c_0 - \frac{1}{2}c_1 \\c_3 &= -\frac{1}{3}c_2 \\c_4 &= -\frac{1}{4}c_3 + \frac{1}{12}c_2 - \frac{1}{24}c_1.\end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = \frac{1}{2}, \quad c_3 = -\frac{1}{6}, \quad c_4 = 0$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = -\frac{1}{2}, \quad c_3 = \frac{1}{6}, \quad c_4 = -\frac{1}{24}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \cdots \quad \text{and} \quad y_2 = x - \frac{1}{2}x^2 + \frac{1}{6}x^3 - \frac{1}{24}x^4 + \cdots.$$

- 35.** The singular points of $(\cos x)y'' + y' + 5y = 0$ are odd integer multiples of $\pi/2$. The distance from 0 to either $\pm\pi/2$ is $\pi/2$. The singular point closest to 1 is $\pi/2$. The distance from 1 to the closest singular point is then $\pi/2 - 1$.

- 36.** Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the first differential equation leads to

$$\begin{aligned}y'' - xy &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=0}^{\infty} c_n x^{n+1}}_{k=n+1} = \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k - \sum_{k=1}^{\infty} c_{k-1}x^k \\&= 2c_2 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - c_{k-1}]x^k = 1.\end{aligned}$$

Thus

$$2c_2 = 1$$

$$(k+2)(k+1)c_{k+2} - c_{k-1} = 0$$

and

$$\begin{aligned}c_2 &= \frac{1}{2} \\c_{k+2} &= \frac{c_{k-1}}{(k+2)(k+1)}, \quad k = 1, 2, 3, \dots\end{aligned}$$

Let c_0 and c_1 be arbitrary and iterate to find

$$\begin{aligned}c_2 &= \frac{1}{2} \\c_3 &= \frac{1}{6}c_0 \\c_4 &= \frac{1}{12}c_1 \\c_5 &= \frac{1}{20}c_2 = \frac{1}{40}\end{aligned}$$

and so on. The solution is

$$\begin{aligned} y &= c_0 + c_1x + \frac{1}{2}x^2 + \frac{1}{6}c_0x^3 + \frac{1}{12}c_1x^4 + \frac{1}{40}c_5 + \cdots \\ &= c_0 \left(1 + \frac{1}{6}x^3 + \cdots \right) + c_1 \left(x + \frac{1}{12}x^4 + \cdots \right) + \frac{1}{2}x^2 + \frac{1}{40}x^5 + \cdots . \end{aligned}$$

Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the second differential equation leads to

$$\begin{aligned} y'' - 4xy' - 4y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} 4nc_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} 4c_n x^n}_{k=n} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k - \sum_{k=1}^{\infty} 4kc_kx^k - \sum_{k=0}^{\infty} 4c_kx^k \\ &= 2c_2 - 4c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - 4(k+1)c_k]x^k \\ &= e^x = 1 + \sum_{k=1}^{\infty} \frac{1}{k!}x^k. \end{aligned}$$

Thus

$$\begin{aligned} 2c_2 - 4c_0 &= 1 \\ (k+2)(k+1)c_{k+2} - 4(k+1)c_k &= \frac{1}{k!} \end{aligned}$$

and

$$\begin{aligned} c_2 &= \frac{1}{2} + 2c_0 \\ c_{k+2} &= \frac{1}{(k+2)!} + \frac{4}{k+2}c_k, \quad k = 1, 2, 3, \dots \end{aligned}$$

Let c_0 and c_1 be arbitrary and iterate to find

$$\begin{aligned} c_2 &= \frac{1}{2} + 2c_0 \\ c_3 &= \frac{1}{3!} + \frac{4}{3}c_1 = \frac{1}{3!} + \frac{4}{3}c_1 \\ c_4 &= \frac{1}{4!} + \frac{4}{4}c_2 = \frac{1}{4!} + \frac{1}{2} + 2c_0 = \frac{13}{4!} + 2c_0 \\ c_5 &= \frac{1}{5!} + \frac{4}{5}c_3 = \frac{1}{5!} + \frac{4}{5 \cdot 3!} + \frac{16}{15}c_1 = \frac{17}{5!} + \frac{16}{15}c_1 \\ c_6 &= \frac{1}{6!} + \frac{4}{6}c_4 = \frac{1}{6!} + \frac{4 \cdot 13}{6 \cdot 4!} + \frac{8}{6}c_0 = \frac{261}{6!} + \frac{4}{3}c_0 \\ c_7 &= \frac{1}{7!} + \frac{4}{7}c_5 = \frac{1}{7!} + \frac{4 \cdot 17}{7 \cdot 5!} + \frac{64}{105}c_1 = \frac{409}{7!} + \frac{64}{105}c_1 \end{aligned}$$

and so on. The solution is

5.1 Solutions About Ordinary Points

$$\begin{aligned}
 y &= c_0 + c_1x + \left(\frac{1}{2} + 2c_0\right)x^2 + \left(\frac{1}{3!} + \frac{4}{3}c_1\right)x^3 + \left(\frac{13}{4!} + 2c_0\right)x^4 + \left(\frac{17}{5!} + \frac{16}{15}c_1\right)x^5 \\
 &\quad + \left(\frac{261}{6!} + \frac{4}{3}c_0\right)x^6 + \left(\frac{409}{7!} + \frac{64}{105}c_1\right)x^7 + \cdots \\
 &= c_0 \left[1 + 2x^2 + 2x^4 + \frac{4}{3}x^6 + \cdots\right] + c_1 \left[x + \frac{4}{3}x^3 + \frac{16}{15}x^5 + \frac{64}{105}x^7 + \cdots\right] \\
 &\quad + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \frac{13}{4!}x^4 + \frac{17}{5!}x^5 + \frac{261}{6!}x^6 + \frac{409}{7!}x^7 + \cdots.
 \end{aligned}$$

37. We identify $P(x) = 0$ and $Q(x) = \sin x/x$. The Taylor series representation for $\sin x/x$ is $1 - x^2/3! + x^4/5! - \cdots$, for $|x| < \infty$. Thus, $Q(x)$ is analytic at $x = 0$ and $x = 0$ is an ordinary point of the differential equation.

38. If $x > 0$ and $y > 0$, then $y'' = -xy < 0$ and the graph of a solution curve is concave down. Thus, whatever portion of a solution curve lies in the first quadrant is concave down. When $x > 0$ and $y < 0$, $y'' = -xy > 0$, so whatever portion of a solution curve lies in the fourth quadrant is concave up.

39. (a) Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 y'' + xy' + y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} + \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\
 &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=1}^{\infty} k c_k x^k + \sum_{k=0}^{\infty} c_k x^k \\
 &= (2c_2 + c_0) + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + (k+1)c_k] x^k = 0.
 \end{aligned}$$

Thus

$$\begin{aligned}
 2c_2 + c_0 &= 0 \\
 (k+2)(k+1)c_{k+2} + (k+1)c_k &= 0
 \end{aligned}$$

and

$$\begin{aligned}
 c_2 &= -\frac{1}{2}c_0 \\
 c_{k+2} &= -\frac{1}{k+2}c_k, \quad k = 1, 2, 3, \dots
 \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned}
 c_2 &= -\frac{1}{2} \\
 c_3 &= c_5 = c_7 = \cdots = 0 \\
 c_4 &= -\frac{1}{4} \left(-\frac{1}{2}\right) = \frac{1}{2^2 \cdot 2} \\
 c_6 &= -\frac{1}{6} \left(\frac{1}{2^2 \cdot 2}\right) = -\frac{1}{2^3 \cdot 3!}
 \end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = c_4 = c_6 = \cdots = 0$$

$$c_3 = -\frac{1}{3} = -\frac{2}{3!}$$

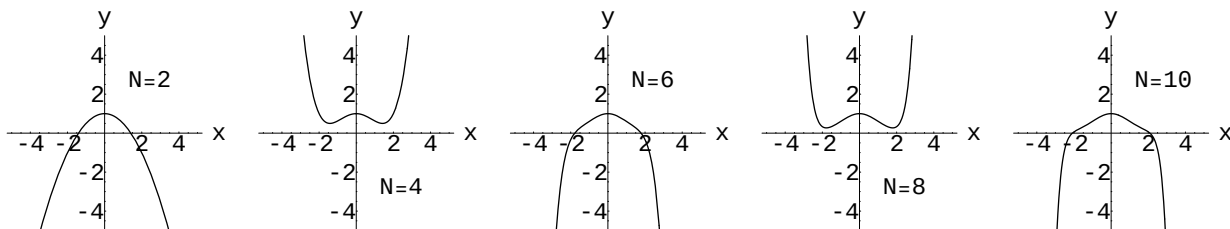
$$c_5 = -\frac{1}{5} \left(-\frac{1}{3} \right) = \frac{1}{5 \cdot 3} = \frac{4 \cdot 2}{5!}$$

$$c_7 = -\frac{1}{7} \left(\frac{4 \cdot 2}{5!} \right) = -\frac{6 \cdot 4 \cdot 2}{7!}$$

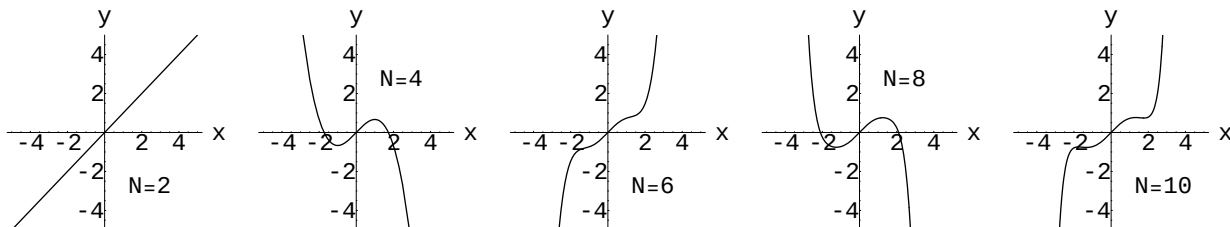
and so on. Thus, two solutions are

$$y_1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{2^k \cdot k!} x^{2k} \quad \text{and} \quad y_2 = \sum_{k=0}^{\infty} \frac{(-1)^k 2^k k!}{(2k+1)!} x^{2k+1}.$$

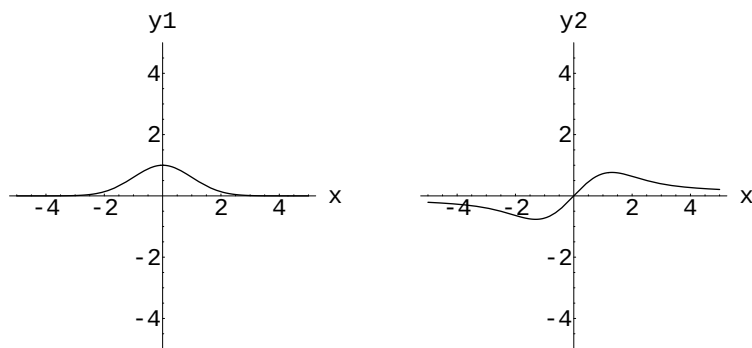
(b) For y_1 , $S_3 = S_2$ and $S_5 = S_4$, so we plot S_2, S_4, S_6, S_8 , and S_{10} .



For y_2 , $S_3 = S_4$ and $S_5 = S_6$, so we plot S_2, S_4, S_6, S_8 , and S_{10} .



(c)



The graphs of y_1 and y_2 obtained from a numerical solver are shown. We see that the partial sum representations indicate the even and odd natures of the solution, but don't really give a very accurate representation of the true solution. Increasing N to about 20 gives a much more accurate representation on $[-4, 4]$.

(d) From $e^x = \sum_{k=0}^{\infty} x^k/k!$ we see that $e^{-x^2/2} = \sum_{k=0}^{\infty} (-x^2/2)^k/k! = \sum_{k=0}^{\infty} (-1)^k x^{2k}/2^k k!$. From (5) of Section 3.2 we have

5.1 Solutions About Ordinary Points

$$\begin{aligned}
 y_2 &= y_1 \int \frac{e^{-\int x dx}}{y_1^2} dx = e^{-x^2/2} \int \frac{e^{-x^2/2}}{(e^{-x^2/2})^2} dx = e^{-x^2/2} \int \frac{e^{-x^2/2}}{e^{-x^2}} dx = e^{-x^2/2} \int e^{x^2/2} dx \\
 &= \sum_{k=0}^{\infty} \frac{(-1)^k}{2^k k!} x^{2k} \int \sum_{k=0}^{\infty} \frac{1}{2^k k!} x^{2k} dx = \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k k!} x^{2k} \right) \left(\sum_{k=0}^{\infty} \int \frac{1}{2^k k!} x^{2k} dx \right) \\
 &= \left(\sum_{k=0}^{\infty} \frac{(-1)^k}{2^k k!} x^{2k} \right) \left(\sum_{k=0}^{\infty} \frac{1}{(2k+1)2^k k!} x^{2k+1} \right) \\
 &= \left(1 - \frac{1}{2}x^2 + \frac{1}{2^2 \cdot 2}x^4 - \frac{1}{2^3 \cdot 3!}x^6 + \cdots \right) \left(x + \frac{1}{3 \cdot 2}x^3 + \frac{1}{5 \cdot 2^2 \cdot 2}x^5 + \frac{1}{7 \cdot 2^3 \cdot 3!}x^7 + \cdots \right) \\
 &= x - \frac{2}{3!}x^3 + \frac{4 \cdot 2}{5!}x^5 - \frac{6 \cdot 4 \cdot 2}{7!}x^7 + \cdots = \sum_{k=0}^{\infty} \frac{(-1)^k 2^k k!}{(2k+1)!} x^{2k+1}.
 \end{aligned}$$

40. (a) We have

$$\begin{aligned}
 y'' + (\cos x)y &= 2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + 42c_7x^5 + \cdots \\
 &\quad + \left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \right) (c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5 + \cdots) \\
 &= (2c_2 + c_0) + (6c_3 + c_1)x + \left(12c_4 + c_2 - \frac{1}{2}c_0 \right)x^2 + \left(20c_5 + c_3 - \frac{1}{2}c_1 \right)x^3 \\
 &\quad + \left(30c_6 + c_4 + \frac{1}{24}c_0 - \frac{1}{2}c_2 \right)x^4 + \left(42c_7 + c_5 + \frac{1}{24}c_1 - \frac{1}{2}c_3 \right)x^5 + \cdots.
 \end{aligned}$$

Then

$$30c_6 + c_4 + \frac{1}{24}c_0 - \frac{1}{2}c_2 = 0 \quad \text{and} \quad 42c_7 + c_5 + \frac{1}{24}c_1 - \frac{1}{2}c_3 = 0,$$

which gives $c_6 = -c_0/80$ and $c_7 = -19c_1/5040$. Thus

$$y_1(x) = 1 - \frac{1}{2}x^2 + \frac{1}{12}x^4 - \frac{1}{80}x^6 + \cdots$$

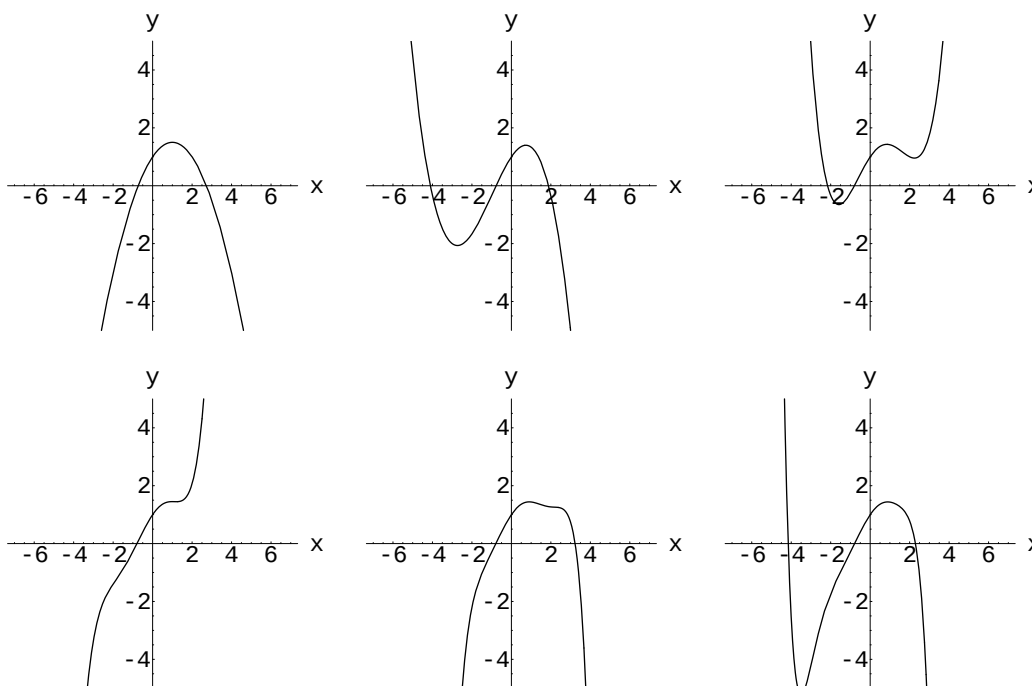
and

$$y_2(x) = x - \frac{1}{6}x^3 + \frac{1}{30}x^5 - \frac{19}{5040}x^7 + \cdots.$$

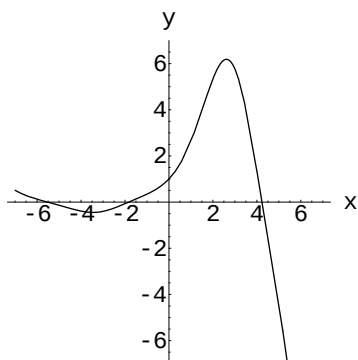
(b) From part (a) the general solution of the differential equation is $y = c_1y_1 + c_2y_2$. Then $y(0) = c_1 + c_2 \cdot 0 = c_1$ and $y'(0) = c_1 \cdot 0 + c_2 = c_2$, so the solution of the initial-value problem is

$$y = y_1 + y_2 = 1 + x - \frac{1}{2}x^2 - \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{30}x^5 - \frac{1}{80}x^6 - \frac{19}{5040}x^7 + \cdots.$$

(c)



(d)



EXERCISES 5.2

Solutions About Singular Points

1. Irregular singular point: $x = 0$
2. Regular singular points: $x = 0, -3$
3. Irregular singular point: $x = 3$; regular singular point: $x = -3$
4. Irregular singular point: $x = 1$; regular singular point: $x = 0$
5. Regular singular points: $x = 0, \pm 2i$
6. Irregular singular point: $x = 5$; regular singular point: $x = 0$

5.2 Solutions About Singular Points

7. Regular singular points: $x = -3, 2$
8. Regular singular points: $x = 0, \pm i$
9. Irregular singular point: $x = 0$; regular singular points: $x = 2, \pm 5$
10. Irregular singular point: $x = -1$; regular singular points: $x = 0, 3$
11. Writing the differential equation in the form

$$y'' + \frac{5}{x-1}y' + \frac{x}{x+1}y = 0$$

we see that $x_0 = 1$ and $x_0 = -1$ are regular singular points. For $x_0 = 1$ the differential equation can be put in the form

$$(x-1)^2y'' + 5(x-1)y' + \frac{x(x-1)^2}{x+1}y = 0.$$

In this case $p(x) = 5$ and $q(x) = x(x-1)^2/(x+1)$. For $x_0 = -1$ the differential equation can be put in the form

$$(x+1)^2y'' + 5(x+1)\frac{x+1}{x-1}y' + x(x+1)y = 0.$$

In this case $p(x) = (x+1)/(x-1)$ and $q(x) = x(x+1)$.

12. Writing the differential equation in the form

$$y'' + \frac{x+3}{x}y' + 7xy = 0$$

we see that $x_0 = 0$ is a regular singular point. Multiplying by x^2 , the differential equation can be put in the form

$$x^2y'' + x(x+3)y' + 7x^3y = 0.$$

We identify $p(x) = x+3$ and $q(x) = 7x^3$.

13. We identify $P(x) = 5/3x + 1$ and $Q(x) = -1/3x^2$, so that $p(x) = xP(x) = \frac{5}{3} + x$ and $q(x) = x^2Q(x) = -\frac{1}{3}$. Then $a_0 = \frac{5}{3}$, $b_0 = -\frac{1}{3}$, and the indicial equation is

$$r(r-1) + \frac{5}{3}r - \frac{1}{3} = r^2 + \frac{2}{3}r - \frac{1}{3} = \frac{1}{3}(3r^2 + 2r - 1) = \frac{1}{3}(3r-1)(r+1) = 0.$$

The indicial roots are $\frac{1}{3}$ and -1 . Since these do not differ by an integer we expect to find two series solutions using the method of Frobenius.

14. We identify $P(x) = 1/x$ and $Q(x) = 10/x$, so that $p(x) = xP(x) = 1$ and $q(x) = x^2Q(x) = 10x$. Then $a_0 = 1$, $b_0 = 0$, and the indicial equation is

$$r(r-1) + r = r^2 = 0.$$

The indicial roots are 0 and 0. Since these are equal, we expect the method of Frobenius to yield a single series solution.

15. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$2xy'' - y' + 2y = (2r^2 - 3r)c_0x^{r-1} + \sum_{k=1}^{\infty} [2(k+r-1)(k+r)c_k - (k+r)c_k + 2c_{k-1}]x^{k+r-1} = 0,$$

which implies

$$2r^2 - 3r = r(2r-3) = 0$$

and

$$(k+r)(2k+2r-3)c_k + 2c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 3/2$. For $r = 0$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{k(2k-3)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = 2c_0, \quad c_2 = -2c_0, \quad c_3 = \frac{4}{9}c_0,$$

and so on. For $r = 3/2$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{(2k+3)k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{2}{5}c_0, \quad c_2 = \frac{2}{35}c_0, \quad c_3 = -\frac{4}{945}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 \left(1 + 2x - 2x^2 + \frac{4}{9}x^3 + \dots \right) + C_2 x^{3/2} \left(1 - \frac{2}{5}x + \frac{2}{35}x^2 - \frac{4}{945}x^3 + \dots \right).$$

16. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 2xy'' + 5y' + xy &= (2r^2 + 3r)c_0 x^{r-1} + (2r^2 + 7r + 5)c_1 x^r \\ &+ \sum_{k=2}^{\infty} [2(k+r)(k+r-1)c_k + 5(k+r)c_k + c_{k-2}]x^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$\begin{aligned} 2r^2 + 3r &= r(2r + 3) = 0, \\ (2r^2 + 7r + 5)c_1 &= 0, \end{aligned}$$

and

$$(k+r)(2k+2r+3)c_k + c_{k-2} = 0.$$

The indicial roots are $r = -3/2$ and $r = 0$, so $c_1 = 0$. For $r = -3/2$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{(2k-3)k}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{1}{2}c_0, \quad c_3 = 0, \quad c_4 = \frac{1}{40}c_0,$$

and so on. For $r = 0$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(2k+3)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{1}{14}c_0, \quad c_3 = 0, \quad c_4 = \frac{1}{616}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 x^{-3/2} \left(1 - \frac{1}{2}x^2 + \frac{1}{40}x^4 + \dots \right) + C_2 \left(1 - \frac{1}{14}x^2 + \frac{1}{616}x^4 + \dots \right).$$

5.2 Solutions About Singular Points

17. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$4xy'' + \frac{1}{2}y' + y = \left(4r^2 - \frac{7}{2}r\right) c_0 x^{r-1} + \sum_{k=1}^{\infty} \left[4(k+r)(k+r-1)c_k + \frac{1}{2}(k+r)c_k + c_{k-1}\right] x^{k+r-1} = 0,$$

which implies

$$4r^2 - \frac{7}{2}r = r \left(4r - \frac{7}{2}\right) = 0$$

and

$$\frac{1}{2}(k+r)(8k+8r-7)c_k + c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 7/8$. For $r = 0$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{k(8k-7)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -2c_0, \quad c_2 = \frac{2}{9}c_0, \quad c_3 = -\frac{4}{459}c_0,$$

and so on. For $r = 7/8$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{(8k+7)k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{2}{15}c_0, \quad c_2 = \frac{2}{345}c_0, \quad c_3 = -\frac{4}{32,085}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 \left(1 - 2x + \frac{2}{9}x^2 - \frac{4}{459}x^3 + \dots\right) + C_2 x^{7/8} \left(1 - \frac{2}{15}x + \frac{2}{345}x^2 - \frac{4}{32,085}x^3 + \dots\right).$$

18. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$2x^2y'' - xy' + (x^2 + 1)y = (2r^2 - 3r + 1)c_0x^r + (2r^2 + r)c_1x^{r+1} + \sum_{k=2}^{\infty} [2(k+r)(k+r-1)c_k - (k+r)c_k + c_k + c_{k-2}]x^{k+r} = 0,$$

which implies

$$2r^2 - 3r + 1 = (2r - 1)(r - 1) = 0, \\ (2r^2 + r)c_1 = 0,$$

and

$$[(k+r)(2k+2r-3) + 1]c_k + c_{k-2} = 0.$$

The indicial roots are $r = 1/2$ and $r = 1$, so $c_1 = 0$. For $r = 1/2$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(2k-1)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{1}{6}c_0, \quad c_3 = 0, \quad c_4 = \frac{1}{168}c_0,$$

and so on. For $r = 1$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(2k+1)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{1}{10}c_0, \quad c_3 = 0, \quad c_4 = \frac{1}{360}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 x^{1/2} \left(1 - \frac{1}{6}x^2 + \frac{1}{168}x^4 + \dots \right) + C_2 x \left(1 - \frac{1}{10}x^2 + \frac{1}{360}x^4 + \dots \right).$$

19. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 3xy'' + (2-x)y' - y &= (3r^2 - r)c_0 x^{r-1} \\ &+ \sum_{k=1}^{\infty} [3(k+r-1)(k+r)c_k + 2(k+r)c_k - (k+r)c_{k-1}]x^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$3r^2 - r = r(3r-1) = 0$$

and

$$(k+r)(3k+3r-1)c_k - (k+r)c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 1/3$. For $r = 0$ the recurrence relation is

$$c_k = \frac{c_{k-1}}{3k-1}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{1}{2}c_0, \quad c_2 = \frac{1}{10}c_0, \quad c_3 = \frac{1}{80}c_0,$$

and so on. For $r = 1/3$ the recurrence relation is

$$c_k = \frac{c_{k-1}}{3k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{1}{3}c_0, \quad c_2 = \frac{1}{18}c_0, \quad c_3 = \frac{1}{162}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 \left(1 + \frac{1}{2}x + \frac{1}{10}x^2 + \frac{1}{80}x^3 + \dots \right) + C_2 x^{1/3} \left(1 + \frac{1}{3}x + \frac{1}{18}x^2 + \frac{1}{162}x^3 + \dots \right).$$

20. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} x^2 y'' - \left(x - \frac{2}{9} \right) y &= \left(r^2 - r + \frac{2}{9} \right) c_0 x^r + \sum_{k=1}^{\infty} \left[(k+r)(k+r-1)c_k + \frac{2}{9}c_k - c_{k-1} \right] x^{k+r} \\ &= 0, \end{aligned}$$

which implies

$$r^2 - r + \frac{2}{9} = \left(r - \frac{2}{3} \right) \left(r - \frac{1}{3} \right) = 0$$

and

$$\left[(k+r)(k+r-1) + \frac{2}{9} \right] c_k - c_{k-1} = 0.$$

5.2 Solutions About Singular Points

The indicial roots are $r = 2/3$ and $r = 1/3$. For $r = 2/3$ the recurrence relation is

$$c_k = \frac{3c_{k-1}}{3k^2 + k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{3}{4}c_0, \quad c_2 = \frac{9}{56}c_0, \quad c_3 = \frac{9}{560}c_0,$$

and so on. For $r = 1/3$ the recurrence relation is

$$c_k = \frac{3c_{k-1}}{3k^2 - k}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{3}{2}c_0, \quad c_2 = \frac{9}{20}c_0, \quad c_3 = \frac{9}{160}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 x^{2/3} \left(1 + \frac{3}{4}x + \frac{9}{56}x^2 + \frac{9}{560}x^3 + \dots \right) + C_2 x^{1/3} \left(1 + \frac{3}{2}x + \frac{9}{20}x^2 + \frac{9}{160}x^3 + \dots \right).$$

21. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} 2xy'' - (3+2x)y' + y &= (2r^2 - 5r)c_0 x^{r-1} + \sum_{k=1}^{\infty} [2(k+r)(k+r-1)c_k \\ &\quad - 3(k+r)c_k - 2(k+r-1)c_{k-1} + c_{k-1}]x^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$2r^2 - 5r = r(2r - 5) = 0$$

and

$$(k+r)(2k+2r-5)c_k - (2k+2r-3)c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 5/2$. For $r = 0$ the recurrence relation is

$$c_k = \frac{(2k-3)c_{k-1}}{k(2k-5)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{1}{3}c_0, \quad c_2 = -\frac{1}{6}c_0, \quad c_3 = -\frac{1}{6}c_0,$$

and so on. For $r = 5/2$ the recurrence relation is

$$c_k = \frac{2(k+1)c_{k-1}}{k(2k+5)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = \frac{4}{7}c_0, \quad c_2 = \frac{4}{21}c_0, \quad c_3 = \frac{32}{693}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 \left(1 + \frac{1}{3}x - \frac{1}{6}x^2 - \frac{1}{6}x^3 + \dots \right) + C_2 x^{5/2} \left(1 + \frac{4}{7}x + \frac{4}{21}x^2 + \frac{32}{693}x^3 + \dots \right).$$

22. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} x^2 y'' + x y' + \left(x^2 - \frac{4}{9}\right) y &= \left(r^2 - \frac{4}{9}\right) c_0 x^r + \left(r^2 + 2r + \frac{5}{9}\right) c_1 x^{r+1} \\ &+ \sum_{k=2}^{\infty} \left[(k+r)(k+r-1)c_k + (k+r)c_k - \frac{4}{9}c_k + c_{k-2} \right] x^{k+r} \\ &= 0, \end{aligned}$$

which implies

$$\begin{aligned} r^2 - \frac{4}{9} &= \left(r + \frac{2}{3}\right) \left(r - \frac{2}{3}\right) = 0, \\ \left(r^2 + 2r + \frac{5}{9}\right) c_1 &= 0, \end{aligned}$$

and

$$\left[(k+r)^2 - \frac{4}{9} \right] c_k + c_{k-2} = 0.$$

The indicial roots are $r = -2/3$ and $r = 2/3$, so $c_1 = 0$. For $r = -2/3$ the recurrence relation is

$$c_k = -\frac{9c_{k-2}}{3k(3k-4)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{3}{4}c_0, \quad c_3 = 0, \quad c_4 = \frac{9}{128}c_0,$$

and so on. For $r = 2/3$ the recurrence relation is

$$c_k = -\frac{9c_{k-2}}{3k(3k+4)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = -\frac{3}{20}c_0, \quad c_3 = 0, \quad c_4 = \frac{9}{1,280}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 x^{-2/3} \left(1 - \frac{3}{4}x^2 + \frac{9}{128}x^4 + \dots \right) + C_2 x^{2/3} \left(1 - \frac{3}{20}x^2 + \frac{9}{1,280}x^4 + \dots \right).$$

23. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$9x^2 y'' + 9x^2 y' + 2y = (9r^2 - 9r + 2) c_0 x^r + \sum_{k=1}^{\infty} [9(k+r)(k+r-1)c_k + 2c_k + 9(k+r-1)c_{k-1}] x^{k+r} = 0,$$

which implies

$$9r^2 - 9r + 2 = (3r-1)(3r-2) = 0$$

and

$$[9(k+r)(k+r-1) + 2]c_k + 9(k+r-1)c_{k-1} = 0.$$

The indicial roots are $r = 1/3$ and $r = 2/3$. For $r = 1/3$ the recurrence relation is

$$c_k = -\frac{(3k-2)c_{k-1}}{k(3k-1)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{1}{2}c_0, \quad c_2 = \frac{1}{5}c_0, \quad c_3 = -\frac{7}{120}c_0,$$

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and so on. For $r = 2/3$ the recurrence relation is

$$c_k = -\frac{(3k-1)c_{k-1}}{k(3k+1)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{1}{2}c_0, \quad c_2 = \frac{5}{28}c_0, \quad c_3 = -\frac{1}{21}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 x^{1/3} \left(1 - \frac{1}{2}x + \frac{1}{5}x^2 - \frac{7}{120}x^3 + \dots \right) + C_2 x^{2/3} \left(1 - \frac{1}{2}x + \frac{5}{28}x^2 - \frac{1}{21}x^3 + \dots \right).$$

- 24.** Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$2x^2 y'' + 3xy' + (2x-1)y = (2r^2 + r - 1)c_0 x^r + \sum_{k=1}^{\infty} [2(k+r)(k+r-1)c_k + 3(k+r)c_k - c_k + 2c_{k-1}]x^{k+r} = 0,$$

which implies

$$2r^2 + r - 1 = (2r-1)(r+1) = 0$$

and

$$[(k+r)(2k+2r+1) - 1]c_k + 2c_{k-1} = 0.$$

The indicial roots are $r = -1$ and $r = 1/2$. For $r = -1$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{k(2k-3)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = 2c_0, \quad c_2 = -2c_0, \quad c_3 = \frac{4}{9}c_0,$$

and so on. For $r = 1/2$ the recurrence relation is

$$c_k = -\frac{2c_{k-1}}{k(2k+3)}, \quad k = 1, 2, 3, \dots,$$

and

$$c_1 = -\frac{2}{5}c_0, \quad c_2 = \frac{2}{35}c_0, \quad c_3 = -\frac{4}{945}c_0,$$

and so on. The general solution on $(0, \infty)$ is

$$y = C_1 x^{-1} \left(1 + 2x - 2x^2 + \frac{4}{9}x^3 + \dots \right) + C_2 x^{1/2} \left(1 - \frac{2}{5}x + \frac{2}{35}x^2 - \frac{4}{945}x^3 + \dots \right).$$

- 25.** Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$xy'' + 2y' - xy = (r^2 + r)c_0 x^{r-1} + (r^2 + 3r + 2)c_1 x^r + \sum_{k=2}^{\infty} [(k+r)(k+r-1)c_k + 2(k+r)c_k - c_{k-2}]x^{k+r-1} = 0,$$

which implies

$$r^2 + r = r(r+1) = 0,$$

$$(r^2 + 3r + 2)c_1 = 0,$$

and

$$(k+r)(k+r+1)c_k - c_{k-2} = 0.$$

The indicial roots are $r_1 = 0$ and $r_2 = -1$, so $c_1 = 0$. For $r_1 = 0$ the recurrence relation is

$$c_k = \frac{c_{k-2}}{k(k+1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned} c_2 &= \frac{1}{3!}c_0 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{1}{5!}c_0 \\ c_{2n} &= \frac{1}{(2n+1)!}c_0. \end{aligned}$$

For $r_2 = -1$ the recurrence relation is

$$c_k = \frac{c_{k-2}}{k(k-1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned} c_2 &= \frac{1}{2!}c_0 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{1}{4!}c_0 \\ c_{2n} &= \frac{1}{(2n)!}c_0. \end{aligned}$$

The general solution on $(0, \infty)$ is

$$\begin{aligned} y &= C_1 \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n} + C_2 x^{-1} \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n} \\ &= \frac{1}{x} \left[C_1 \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1} + C_2 \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n} \right] \\ &= \frac{1}{x} [C_1 \sinh x + C_2 \cosh x]. \end{aligned}$$

26. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} x^2 y'' + x y' + \left(x^2 - \frac{1}{4}\right) y &= \left(r^2 - \frac{1}{4}\right) c_0 x^r + \left(r^2 + 2r + \frac{3}{4}\right) c_1 x^{r+1} \\ &\quad + \sum_{k=2}^{\infty} \left[(k+r)(k+r-1)c_k + (k+r)c_k - \frac{1}{4}c_k + c_{k-2} \right] x^{k+r} \\ &= 0, \end{aligned}$$

which implies

$$\begin{aligned} r^2 - \frac{1}{4} &= \left(r - \frac{1}{2}\right) \left(r + \frac{1}{2}\right) = 0, \\ \left(r^2 + 2r + \frac{3}{4}\right) c_1 &= 0, \end{aligned}$$

and

$$\left[(k+r)^2 - \frac{1}{4} \right] c_k + c_{k-2} = 0.$$

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The indicial roots are $r_1 = 1/2$ and $r_2 = -1/2$, so $c_1 = 0$. For $r_1 = 1/2$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(k+1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned} c_2 &= -\frac{1}{3!}c_0 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{1}{5!}c_0 \\ c_{2n} &= \frac{(-1)^n}{(2n+1)!}c_0. \end{aligned}$$

For $r_2 = -1/2$ the recurrence relation is

$$c_k = -\frac{c_{k-2}}{k(k-1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned} c_2 &= -\frac{1}{2!}c_0 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{1}{4!}c_0 \\ c_{2n} &= \frac{(-1)^n}{(2n)!}c_0. \end{aligned}$$

The general solution on $(0, \infty)$ is

$$\begin{aligned} y &= C_1 x^{1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n} + C_2 x^{-1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \\ &= C_1 x^{-1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} + C_2 x^{-1/2} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \\ &= x^{-1/2} [C_1 \sin x + C_2 \cos x]. \end{aligned}$$

27. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$xy'' - xy' + y = (r^2 - r) c_0 x^{r-1} + \sum_{k=0}^{\infty} [(k+r+1)(k+r)c_{k+1} - (k+r)c_k + c_k] x^{k+r} = 0$$

which implies

$$r^2 - r = r(r-1) = 0$$

and

$$(k+r+1)(k+r)c_{k+1} - (k+r-1)c_k = 0.$$

The indicial roots are $r_1 = 1$ and $r_2 = 0$. For $r_1 = 1$ the recurrence relation is

$$c_{k+1} = \frac{kc_k}{(k+2)(k+1)}, \quad k = 0, 1, 2, \dots,$$

and one solution is $y_1 = c_0 x$. A second solution is

$$\begin{aligned} y_2 &= x \int \frac{e^{-\int -1 dx}}{x^2} dx = x \int \frac{e^x}{x^2} dx = x \int \frac{1}{x^2} \left(1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \cdots \right) dx \\ &= x \int \left(\frac{1}{x^2} + \frac{1}{x} + \frac{1}{2} + \frac{1}{3!}x + \frac{1}{4!}x^2 + \cdots \right) dx = x \left[-\frac{1}{x} + \ln x + \frac{1}{2}x + \frac{1}{12}x^2 + \frac{1}{72}x^3 + \cdots \right] \\ &= x \ln x - 1 + \frac{1}{2}x^2 + \frac{1}{12}x^3 + \frac{1}{72}x^4 + \cdots \end{aligned}$$

The general solution on $(0, \infty)$ is

$$y = C_1 x + C_2 y_2(x).$$

28. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} y'' + \frac{3}{x}y' - 2y &= (r^2 + 2r)c_0 x^{r-2} + (r^2 + 4r + 3)c_1 x^{r-1} \\ &\quad + \sum_{k=2}^{\infty} [(k+r)(k+r-1)c_k + 3(k+r)c_k - 2c_{k-2}] x^{k+r-2} \\ &= 0, \end{aligned}$$

which implies

$$r^2 + 2r = r(r+2) = 0$$

$$(r^2 + 4r + 3)c_1 = 0$$

$$(k+r)(k+r+2)c_k - 2c_{k-2} = 0.$$

The indicial roots are $r_1 = 0$ and $r_2 = -2$, so $c_1 = 0$. For $r_1 = 0$ the recurrence relation is

$$c_k = \frac{2c_{k-2}}{k(k+2)}, \quad k = 2, 3, 4, \dots,$$

and

$$c_2 = \frac{1}{4}c_0$$

$$c_3 = c_5 = c_7 = \cdots = 0$$

$$c_4 = \frac{1}{48}c_0$$

$$c_6 = \frac{1}{1,152}c_0.$$

The result is

$$y_1 = c_0 \left(1 + \frac{1}{4}x^2 + \frac{1}{48}x^4 + \frac{1}{1,152}x^6 + \cdots \right).$$

A second solution is

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{-\int (3/x) dx}}{y_1^2} dx = y_1 \int \frac{dx}{x^3 (1 + \frac{1}{4}x^2 + \frac{1}{48}x^4 + \cdots)^2} \\ &= y_1 \int \frac{dx}{x^3 (1 + \frac{1}{2}x^2 + \frac{5}{48}x^4 + \frac{7}{576}x^6 + \cdots)} = y_1 \int \frac{1}{x^3} \left(1 - \frac{1}{2}x^2 + \frac{7}{48}x^4 + \frac{19}{576}x^6 + \cdots \right) dx \\ &= y_1 \int \left(\frac{1}{x^3} - \frac{1}{2x} + \frac{7}{48}x - \frac{19}{576}x^3 + \cdots \right) dx = y_1 \left[-\frac{1}{2x^2} - \frac{1}{2} \ln x + \frac{7}{96}x^2 - \frac{19}{2,304}x^4 + \cdots \right] \\ &= -\frac{1}{2}y_1 \ln x + y \left[-\frac{1}{2x^2} + \frac{7}{96}x^2 - \frac{19}{2,304}x^4 + \cdots \right]. \end{aligned}$$

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The general solution on $(0, \infty)$ is

$$y = C_1 y_1(x) + C_2 y_2(x).$$

29. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$xy'' + (1-x)y' - y = r^2 c_0 x^{r-1} + \sum_{k=1}^{\infty} [(k+r)(k+r-1)c_k + (k+r)c_k - (k+r)c_{k-1}] x^{k+r-1} = 0,$$

which implies $r^2 = 0$ and

$$(k+r)^2 c_k - (k+r)c_{k-1} = 0.$$

The indicial roots are $r_1 = r_2 = 0$ and the recurrence relation is

$$c_k = \frac{c_{k-1}}{k}, \quad k = 1, 2, 3, \dots$$

One solution is

$$y_1 = c_0 \left(1 + x + \frac{1}{2}x^2 + \frac{1}{3!}x^3 + \dots \right) = c_0 e^x.$$

A second solution is

$$\begin{aligned} y_2 &= y_1 \int \frac{e^{-\int (1/x-1)dx}}{e^{2x}} dx = e^x \int \frac{e^x/x}{e^{2x}} dx = e^x \int \frac{1}{x} e^{-x} dx \\ &= e^x \int \frac{1}{x} \left(1 - x + \frac{1}{2}x^2 - \frac{1}{3!}x^3 + \dots \right) dx = e^x \int \left(\frac{1}{x} - 1 + \frac{1}{2}x - \frac{1}{3!}x^2 + \dots \right) dx \\ &= e^x \left[\ln x - x + \frac{1}{2 \cdot 2}x^2 - \frac{1}{3 \cdot 3!}x^3 + \dots \right] = e^x \ln x - e^x \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n \cdot n!} x^n. \end{aligned}$$

The general solution on $(0, \infty)$ is

$$y = C_1 e^x + C_2 e^x \left(\ln x - \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n \cdot n!} x^n \right).$$

30. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$xy'' + y' + y = r^2 c_0 x^{r-1} + \sum_{k=1}^{\infty} [(k+r)(k+r-1)c_k + (k+r)c_k + c_{k-1}] x^{k+r-1} = 0$$

which implies $r^2 = 0$ and

$$(k+r)^2 c_k + c_{k-1} = 0.$$

The indicial roots are $r_1 = r_2 = 0$ and the recurrence relation is

$$c_k = -\frac{c_{k-1}}{k^2}, \quad k = 1, 2, 3, \dots$$

One solution is

$$y_1 = c_0 \left(1 - x + \frac{1}{2^2}x^2 - \frac{1}{(3!)^2}x^3 + \frac{1}{(4!)^2}x^4 - \dots \right) = c_0 \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} x^n.$$

A second solution is

$$\begin{aligned}
 y_2 &= y_1 \int \frac{e^{-\int (1/x) dx}}{y_1^2} dx = y_1 \int \frac{dx}{x \left(1 - x + \frac{1}{4}x^2 - \frac{1}{36}x^3 + \dots\right)^2} \\
 &= y_1 \int \frac{dx}{x \left(1 - 2x + \frac{3}{2}x^2 - \frac{5}{9}x^3 + \frac{35}{288}x^4 - \dots\right)} \\
 &= y_1 \int \frac{1}{x} \left(1 + 2x + \frac{5}{2}x^2 + \frac{23}{9}x^3 + \frac{677}{288}x^4 + \dots\right) dx \\
 &= y_1 \int \left(\frac{1}{x} + 2 + \frac{5}{2}x + \frac{23}{9}x^2 + \frac{677}{288}x^3 + \dots\right) dx \\
 &= y_1 \left[\ln x + 2x + \frac{5}{4}x^2 + \frac{23}{27}x^3 + \frac{677}{1,152}x^4 + \dots\right] \\
 &= y_1 \ln x + y_1 \left(2x + \frac{5}{4}x^2 + \frac{23}{27}x^3 + \frac{677}{1,152}x^4 + \dots\right).
 \end{aligned}$$

The general solution on $(0, \infty)$ is

$$y = C_1 y_1(x) + C_2 y_2(x).$$

31. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned}
 xy'' + (x-6)y' - 3y &= (r^2 - 7r)c_0 x^{r-1} + \sum_{k=1}^{\infty} [(k+r)(k+r-1)c_k + (k+r-1)c_{k-1} \\
 &\quad - 6(k+r)c_k - 3c_{k-1}]x^{k+r-1} = 0,
 \end{aligned}$$

which implies

$$r^2 - 7r = r(r-7) = 0$$

and

$$(k+r)(k+r-7)c_k + (k+r-4)c_{k-1} = 0.$$

The indicial roots are $r_1 = 7$ and $r_2 = 0$. For $r_1 = 7$ the recurrence relation is

$$(k+7)kc_k + (k+3)c_{k-1} = 0, \quad k = 1, 2, 3, \dots,$$

or

$$c_k = -\frac{k+3}{k(k+7)} c_{k-1}, \quad k = 1, 2, 3, \dots$$

Taking $c_0 \neq 0$ we obtain

$$c_1 = -\frac{1}{2}c_0$$

$$c_2 = \frac{5}{18}c_0$$

$$c_3 = -\frac{1}{6}c_0,$$

and so on. Thus, the indicial root $r_1 = 7$ yields a single solution. Now, for $r_2 = 0$ the recurrence relation is

$$k(k-7)c_k + (k-4)c_{k-1} = 0, \quad k = 1, 2, 3, \dots$$

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Then

$$\begin{aligned}
 -6c_1 - 3c_0 &= 0 \\
 -10c_2 - 2c_1 &= 0 \\
 -12c_3 - c_2 &= 0 \\
 -12c_4 + 0c_3 &= 0 \implies c_4 = 0 \\
 -10c_5 + c_4 &= 0 \implies c_5 = 0 \\
 -6c_6 + 2c_5 &= 0 \implies c_6 = 0 \\
 0c_7 + 3c_6 &= 0 \implies c_7 \text{ is arbitrary}
 \end{aligned}$$

and

$$c_k = -\frac{k-4}{k(k-7)} c_{k-1}, \quad k = 8, 9, 10, \dots$$

Taking $c_0 \neq 0$ and $c_7 = 0$ we obtain

$$\begin{aligned}
 c_1 &= -\frac{1}{2}c_0 \\
 c_2 &= \frac{1}{10}c_0 \\
 c_3 &= -\frac{1}{120}c_0 \\
 c_4 &= c_5 = c_6 = \dots = 0.
 \end{aligned}$$

Taking $c_0 = 0$ and $c_7 \neq 0$ we obtain

$$\begin{aligned}
 c_1 &= c_2 = c_3 = c_4 = c_5 = c_6 = 0 \\
 c_8 &= -\frac{1}{2}c_7 \\
 c_9 &= \frac{5}{36}c_7 \\
 c_{10} &= -\frac{1}{36}c_7,
 \end{aligned}$$

and so on. In this case we obtain the two solutions

$$y_1 = 1 - \frac{1}{2}x + \frac{1}{10}x^2 - \frac{1}{120}x^3 \quad \text{and} \quad y_2 = x^7 - \frac{1}{2}x^8 + \frac{5}{36}x^9 - \frac{1}{36}x^{10} + \dots$$

32. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned}
 &x(x-1)y'' + 3y' - 2y \\
 &= (4r - r^2) c_0 x^{r-1} + \sum_{k=1}^{\infty} [(k+r-1)(k+r-12)c_{k-1} - (k+r)(k+r-1)c_k + 3(k+r)c_k - 2c_{k-1}] x^{k+r-1} \\
 &= 0,
 \end{aligned}$$

which implies

$$4r - r^2 = r(4 - r) = 0$$

and

$$-(k+r)(k+r-4)c_k + [(k+r-1)(k+r-2) - 2]c_{k-1} = 0.$$

The indicial roots are $r_1 = 4$ and $r_2 = 0$. For $r_1 = 4$ the recurrence relation is

$$-(k+4)kc_k + [(k+3)(k+2) - 2]c_{k-1} = 0$$

or

$$c_k = \frac{k+1}{k} c_{k-1}, \quad k = 1, 2, 3, \dots$$

Taking $c_0 \neq 0$ we obtain

$$c_1 = 2c_0$$

$$c_2 = 3c_0$$

$$c_3 = 4c_0,$$

and so on. Thus, the indicial root $r_1 = 4$ yields a single solution. For $r_2 = 0$ the recurrence relation is

$$-k(k-4)c_k + k(k-3)c_{k-1} = 0, \quad k = 1, 2, 3, \dots,$$

or

$$-(k-4)c_k + (k-3)c_{k-1} = 0, \quad k = 1, 2, 3, \dots$$

Then

$$3c_1 - 2c_0 = 0$$

$$2c_2 - c_1 = 0$$

$$c_3 + 0c_2 = 0 \Rightarrow c_3 = 0$$

$$0c_4 + c_3 = 0 \Rightarrow c_4 \text{ is arbitrary}$$

and

$$c_k = \frac{(k-3)c_{k-1}}{k-4}, \quad k = 5, 6, 7, \dots$$

Taking $c_0 \neq 0$ and $c_4 = 0$ we obtain

$$c_1 = \frac{2}{3}c_0$$

$$c_2 = \frac{1}{3}c_0$$

$$c_3 = c_4 = c_5 = \dots = 0.$$

Taking $c_0 = 0$ and $c_4 \neq 0$ we obtain

$$c_1 = c_2 = c_3 = 0$$

$$c_5 = 2c_4$$

$$c_6 = 3c_4$$

$$c_7 = 4c_4,$$

and so on. In this case we obtain the two solutions

$$y_1 = 1 + \frac{2}{3}x + \frac{1}{3}x^2 \quad \text{and} \quad y_2 = x^4 + 2x^5 + 3x^6 + 4x^7 + \dots$$

33. (a) From $t = 1/x$ we have $dt/dx = -1/x^2 = -t^2$. Then

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = -t^2 \frac{dy}{dt}$$

and

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(-t^2 \frac{dy}{dt} \right) = -t^2 \frac{d^2y}{dt^2} \frac{dt}{dx} - \frac{dy}{dt} \left(2t \frac{dt}{dx} \right) = t^4 \frac{d^2y}{dt^2} + 2t^3 \frac{dy}{dt}.$$

Now

$$x^4 \frac{d^2y}{dx^2} + \lambda y = \frac{1}{t^4} \left(t^4 \frac{d^2y}{dt^2} + 2t^3 \frac{dy}{dt} \right) + \lambda y = \frac{d^2y}{dt^2} + \frac{2}{t} \frac{dy}{dt} + \lambda y = 0$$

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becomes

$$t \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + \lambda t y = 0.$$

(b) Substituting $y = \sum_{n=0}^{\infty} c_n t^{n+r}$ into the differential equation and collecting terms, we obtain

$$\begin{aligned} t \frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + \lambda t y &= (r^2 + r)c_0 t^{r-1} + (r^2 + 3r + 2)c_1 t^r \\ &+ \sum_{k=2}^{\infty} [(k+r)(k+r-1)c_k + 2(k+r)c_k + \lambda c_{k-2}] t^{k+r-1} \\ &= 0, \end{aligned}$$

which implies

$$\begin{aligned} r^2 + r &= r(r+1) = 0, \\ (r^2 + 3r + 2)c_1 &= 0, \end{aligned}$$

and

$$(k+r)(k+r+1)c_k + \lambda c_{k-2} = 0.$$

The indicial roots are $r_1 = 0$ and $r_2 = -1$, so $c_1 = 0$. For $r_1 = 0$ the recurrence relation is

$$c_k = -\frac{\lambda c_{k-2}}{k(k+1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned} c_2 &= -\frac{\lambda}{3!}c_0 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{\lambda^2}{5!}c_0 \\ &\vdots \\ c_{2n} &= (-1)^n \frac{\lambda^n}{(2n+1)!}c_0. \end{aligned}$$

For $r_2 = -1$ the recurrence relation is

$$c_k = -\frac{\lambda c_{k-2}}{k(k-1)}, \quad k = 2, 3, 4, \dots,$$

and

$$\begin{aligned} c_2 &= -\frac{\lambda}{2!}c_0 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{\lambda^2}{4!}c_0 \\ &\vdots \\ c_{2n} &= (-1)^n \frac{\lambda^n}{(2n)!}c_0. \end{aligned}$$

The general solution on $(0, \infty)$ is

$$\begin{aligned} y(t) &= c_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (\sqrt{\lambda} t)^{2n} + c_2 t^{-1} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (\sqrt{\lambda} t)^{2n} \\ &= \frac{1}{t} \left[C_1 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (\sqrt{\lambda} t)^{2n+1} + C_2 \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (\sqrt{\lambda} t)^{2n} \right] \\ &= \frac{1}{t} [C_1 \sin \sqrt{\lambda} t + C_2 \cos \sqrt{\lambda} t]. \end{aligned}$$

(c) Using $t = 1/x$, the solution of the original equation is

$$y(x) = C_1 x \sin \frac{\sqrt{\lambda}}{x} + C_2 x \cos \frac{\sqrt{\lambda}}{x}.$$

34. (a) From the boundary conditions $y(a) = 0$, $y(b) = 0$ we find

$$\begin{aligned} C_1 \sin \frac{\sqrt{\lambda}}{a} + C_2 \cos \frac{\sqrt{\lambda}}{a} &= 0 \\ C_1 \sin \frac{\sqrt{\lambda}}{b} + C_2 \cos \frac{\sqrt{\lambda}}{b} &= 0. \end{aligned}$$

Since this is a homogeneous system of linear equations, it will have nontrivial solutions for C_1 and C_2 if

$$\begin{aligned} \begin{vmatrix} \sin \frac{\sqrt{\lambda}}{a} & \cos \frac{\sqrt{\lambda}}{a} \\ \sin \frac{\sqrt{\lambda}}{b} & \cos \frac{\sqrt{\lambda}}{b} \end{vmatrix} &= \sin \frac{\sqrt{\lambda}}{a} \cos \frac{\sqrt{\lambda}}{b} - \cos \frac{\sqrt{\lambda}}{a} \sin \frac{\sqrt{\lambda}}{b} \\ &= \sin \left(\frac{\sqrt{\lambda}}{a} - \frac{\sqrt{\lambda}}{b} \right) = \sin \left(\sqrt{\lambda} \frac{b-a}{ab} \right) = 0. \end{aligned}$$

This will be the case if

$$\sqrt{\lambda} \left(\frac{b-a}{ab} \right) = n\pi \quad \text{or} \quad \sqrt{\lambda} = \frac{n\pi ab}{b-a} = \frac{n\pi ab}{L}, \quad n = 1, 2, \dots,$$

or, if

$$\lambda_n = \frac{n^2 \pi^2 a^2 b^2}{L^2} = \frac{P_n b^4}{EI}.$$

The critical loads are then $P_n = n^2 \pi^2 (a/b)^2 EI_0 / L^2$. Using $C_2 = -C_1 \sin(\sqrt{\lambda}/a) / \cos(\sqrt{\lambda}/a)$ we have

$$\begin{aligned} y &= C_1 x \left[\sin \frac{\sqrt{\lambda}}{x} - \frac{\sin(\sqrt{\lambda}/a)}{\cos(\sqrt{\lambda}/a)} \cos \frac{\sqrt{\lambda}}{x} \right] \\ &= C_3 x \left[\sin \frac{\sqrt{\lambda}}{x} \cos \frac{\sqrt{\lambda}}{a} - \cos \frac{\sqrt{\lambda}}{x} \sin \frac{\sqrt{\lambda}}{a} \right] \\ &= C_3 x \sin \sqrt{\lambda} \left(\frac{1}{x} - \frac{1}{a} \right), \end{aligned}$$

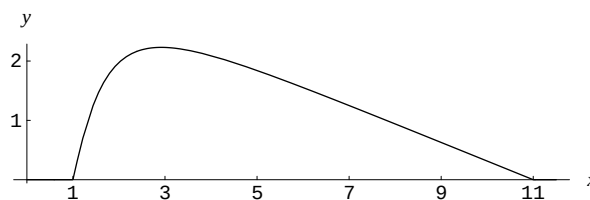
and

$$y_n(x) = C_3 x \sin \frac{n\pi ab}{L} \left(\frac{1}{x} - \frac{1}{a} \right) = C_3 x \sin \frac{n\pi ab}{La} \left(\frac{a}{x} - 1 \right) = C_4 x \sin \frac{n\pi ab}{L} \left(1 - \frac{a}{x} \right).$$

5.2 Solutions About Singular Points

- (b) When $n = 1$, $b = 11$, and $a = 1$, we have,
for $C_4 = 1$,

$$y_1(x) = x \sin 1.1\pi \left(1 - \frac{1}{x}\right).$$



35. Express the differential equation in standard form:

$$y''' + P(x)y'' + Q(x)y' + R(x)y = 0.$$

Suppose x_0 is a singular point of the differential equation. Then we say that x_0 is a regular singular point if $(x - x_0)P(x)$, $(x - x_0)^2Q(x)$, and $(x - x_0)^3R(x)$ are analytic at $x = x_0$.

36. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the first differential equation and collecting terms, we obtain

$$x^3 y'' + y = c_0 x^r + \sum_{k=1}^{\infty} [c_k + (k+r-1)(k+r-2)c_{k-1}]x^{k+r} = 0.$$

It follows that $c_0 = 0$ and

$$c_k = -(k+r-1)(k+r-2)c_{k-1}.$$

The only solution we obtain is $y(x) = 0$.

Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the second differential equation and collecting terms, we obtain

$$x^2 y'' + (3x-1)y' + y = -rc_0 + \sum_{k=0}^{\infty} [(k+r+1)^2 c_k - (k+r+1)c_{k+1}]x^{k+r} = 0,$$

which implies

$$-rc_0 = 0$$

$$(k+r+1)^2 c_k - (k+r+1)c_{k+1} = 0.$$

If $c_0 = 0$, then the solution of the differential equation is $y = 0$. Thus, we take $r = 0$, from which we obtain

$$c_{k+1} = (k+1)c_k, \quad k = 0, 1, 2, \dots$$

Letting $c_0 = 1$ we get $c_1 = 2$, $c_2 = 3!$, $c_3 = 4!$, and so on. The solution of the differential equation is then $y = \sum_{n=0}^{\infty} (n+1)!x^n$, which converges only at $x = 0$.

37. We write the differential equation in the form $x^2 y'' + (b/a)xy' + (c/a)y = 0$ and identify $a_0 = b/a$ and $b_0 = c/a$ as in (12) in the text. Then the indicial equation is

$$r(r-1) + \frac{b}{a}r + \frac{c}{a} = 0 \quad \text{or} \quad ar^2 + (b-a)r + c = 0,$$

which is also the auxiliary equation of $ax^2 y'' + bxy' + cy = 0$.

EXERCISES 5.3

Special Functions

1. Since $\nu^2 = 1/9$ the general solution is $y = c_1 J_{1/3}(x) + c_2 J_{-1/3}(x)$.
2. Since $\nu^2 = 1$ the general solution is $y = c_1 J_1(x) + c_2 Y_1(x)$.
3. Since $\nu^2 = 25/4$ the general solution is $y = c_1 J_{5/2}(x) + c_2 J_{-5/2}(x)$.
4. Since $\nu^2 = 1/16$ the general solution is $y = c_1 J_{1/4}(x) + c_2 J_{-1/4}(x)$.
5. Since $\nu^2 = 0$ the general solution is $y = c_1 J_0(x) + c_2 Y_0(x)$.
6. Since $\nu^2 = 4$ the general solution is $y = c_1 J_2(x) + c_2 Y_2(x)$.
7. We identify $\alpha = 3$ and $\nu = 2$. Then the general solution is $y = c_1 J_2(3x) + c_2 Y_2(3x)$.
8. We identify $\alpha = 6$ and $\nu = \frac{1}{2}$. Then the general solution is $y = c_1 J_{1/2}(6x) + c_2 J_{-1/2}(6x)$.
9. We identify $\alpha = 5$ and $\nu = \frac{2}{3}$. Then the general solution is $y = c_1 J_{2/3}(5x) + c_2 J_{-2/3}(5x)$.
10. We identify $\alpha = \sqrt{2}$ and $\nu = 8$. Then the general solution is $y = c_1 J_8(\sqrt{2}x) + c_2 Y_8(\sqrt{2}x)$.
11. If $y = x^{-1/2}v(x)$ then

$$y' = x^{-1/2}v'(x) - \frac{1}{2}x^{-3/2}v(x),$$

$$y'' = x^{-1/2}v''(x) - x^{-3/2}v'(x) + \frac{3}{4}x^{-5/2}v(x),$$

and

$$x^2 y'' + 2xy' + \alpha^2 x^2 y = x^{3/2}v''(x) + x^{1/2}v'(x) + \left(\alpha^2 x^{3/2} - \frac{1}{4}x^{-1/2}\right)v(x) = 0.$$

Multiplying by $x^{1/2}$ we obtain

$$x^2 v''(x) + xv'(x) + \left(\alpha^2 x^2 - \frac{1}{4}\right)v(x) = 0,$$

whose solution is $v = c_1 J_{1/2}(\alpha x) + c_2 J_{-1/2}(\alpha x)$. Then $y = c_1 x^{-1/2} J_{1/2}(\alpha x) + c_2 x^{-1/2} J_{-1/2}(\alpha x)$.

12. If $y = \sqrt{x}v(x)$ then

$$y' = x^{1/2}v'(x) + \frac{1}{2}x^{-1/2}v(x)$$

$$y'' = x^{1/2}v''(x) + x^{-1/2}v'(x) - \frac{1}{4}x^{-3/2}v(x)$$

and

$$x^2 y'' + \left(\alpha^2 x^2 - \nu^2 + \frac{1}{4}\right)y = x^{5/2}v''(x) + x^{3/2}v'(x) - \frac{1}{4}x^{1/2}v(x) + \left(\alpha^2 x^2 - \nu^2 + \frac{1}{4}\right)x^{1/2}v(x)$$

$$= x^{5/2}v''(x) + x^{3/2}v'(x) + (\alpha^2 x^{5/2} - \nu^2 x^{1/2})v(x) = 0.$$

Multiplying by $x^{-1/2}$ we obtain

$$x^2 v''(x) + xv'(x) + (\alpha^2 x^2 - \nu^2)v(x) = 0,$$

whose solution is $v(x) = c_1 J_\nu(\alpha x) + c_2 Y_\nu(\alpha x)$. Then $y = c_1 \sqrt{x} J_\nu(\alpha x) + c_2 \sqrt{x} Y_\nu(\alpha x)$.

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13. Write the differential equation in the form $y'' + (2/x)y' + (4/x)y = 0$. This is the form of (18) in the text with $a = -\frac{1}{2}$, $c = \frac{1}{2}$, $b = 4$, and $p = 1$, so, by (19) in the text, the general solution is

$$y = x^{-1/2}[c_1 J_1(4x^{1/2}) + c_2 Y_1(4x^{1/2})].$$

14. Write the differential equation in the form $y'' + (3/x)y' + y = 0$. This is the form of (18) in the text with $a = -1$, $c = 1$, $b = 1$, and $p = 1$, so, by (19) in the text, the general solution is

$$y = x^{-1}[c_1 J_1(x) + c_2 Y_1(x)].$$

15. Write the differential equation in the form $y'' - (1/x)y' + y = 0$. This is the form of (18) in the text with $a = 1$, $c = 1$, $b = 1$, and $p = 1$, so, by (19) in the text, the general solution is

$$y = x[c_1 J_1(x) + c_2 Y_1(x)].$$

16. Write the differential equation in the form $y'' - (5/x)y' + y = 0$. This is the form of (18) in the text with $a = 3$, $c = 1$, $b = 1$, and $p = 2$, so, by (19) in the text, the general solution is

$$y = x^3[c_1 J_3(x) + c_2 Y_3(x)].$$

17. Write the differential equation in the form $y'' + (1 - 2/x^2)y = 0$. This is the form of (18) in the text with $a = \frac{1}{2}$, $c = 1$, $b = 1$, and $p = \frac{3}{2}$, so, by (19) in the text, the general solution is

$$y = x^{1/2}[c_1 J_{3/2}(x) + c_2 Y_{3/2}(x)] = x^{1/2}[C_1 J_{3/2}(x) + C_2 J_{-3/2}(x)].$$

18. Write the differential equation in the form $y'' + (4 + 1/4x^2)y = 0$. This is the form of (18) in the text with $a = \frac{1}{2}$, $c = 1$, $b = 2$, and $p = 0$, so, by (19) in the text, the general solution is

$$y = x^{1/2}[c_1 J_0(2x) + c_2 Y_0(2x)].$$

19. Write the differential equation in the form $y'' + (3/x)y' + x^2 y = 0$. This is the form of (18) in the text with $a = -1$, $c = 2$, $b = \frac{1}{2}$, and $p = \frac{1}{2}$, so, by (19) in the text, the general solution is

$$y = x^{-1} \left[c_1 J_{1/2} \left(\frac{1}{2} x^2 \right) + c_2 Y_{1/2} \left(\frac{1}{2} x^2 \right) \right]$$

or

$$y = x^{-1} \left[C_1 J_{1/2} \left(\frac{1}{2} x^2 \right) + C_2 J_{-1/2} \left(\frac{1}{2} x^2 \right) \right].$$

20. Write the differential equation in the form $y'' + (1/x)y' + (\frac{1}{9}x^4 - 4/x^2)y = 0$. This is the form of (18) in the text with $a = 0$, $c = 3$, $b = \frac{1}{9}$, and $p = \frac{2}{3}$, so, by (19) in the text, the general solution is

$$y = c_1 J_{2/3} \left(\frac{1}{9} x^3 \right) + c_2 Y_{2/3} \left(\frac{1}{9} x^3 \right)$$

or

$$y = C_1 J_{2/3} \left(\frac{1}{9} x^3 \right) + C_2 J_{-2/3} \left(\frac{1}{9} x^3 \right).$$

21. Using the fact that $i^2 = -1$, along with the definition of $J_\nu(x)$ in (7) in the text, we have

$$\begin{aligned} I_\nu(x) &= i^{-\nu} J_\nu(ix) = i^{-\nu} \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(1 + \nu + n)} \left(\frac{ix}{2} \right)^{2n+\nu} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(1 + \nu + n)} i^{2n+\nu-\nu} \left(\frac{x}{2} \right)^{2n+\nu} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(1 + \nu + n)} (i^2)^n \left(\frac{x}{2} \right)^{2n+\nu} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{n! \Gamma(1 + \nu + n)} \left(\frac{x}{2} \right)^{2n+\nu} \\ &= \sum_{n=0}^{\infty} \frac{1}{n! \Gamma(1 + \nu + n)} \left(\frac{x}{2} \right)^{2n+\nu}, \end{aligned}$$

which is a real function.

22. (a) The differential equation has the form of (18) in the text with

$$\begin{aligned} 1 - 2a &= 0 \implies a = \frac{1}{2} \\ 2c - 2 &= 2 \implies c = 2 \\ b^2 c^2 &= -\beta^2 c^2 = -1 \implies \beta = \frac{1}{2} \quad \text{and} \quad b = \frac{1}{2}i \\ a^2 - p^2 c^2 &= 0 \implies p = \frac{1}{4}. \end{aligned}$$

Then, by (19) in the text,

$$y = x^{1/2} \left[c_1 J_{1/4} \left(\frac{1}{2} i x^2 \right) + c_2 J_{-1/4} \left(\frac{1}{2} i x^2 \right) \right].$$

In terms of real functions the general solution can be written

$$y = x^{1/2} \left[C_1 I_{1/4} \left(\frac{1}{2} x^2 \right) + C_2 K_{1/4} \left(\frac{1}{2} x^2 \right) \right].$$

(b) Write the differential equation in the form $y'' + (1/x)y' - 7x^2y = 0$. This is the form of (18) in the text with

$$\begin{aligned} 1 - 2a &= 1 \implies a = 0 \\ 2c - 2 &= 2 \implies c = 2 \\ b^2 c^2 &= -\beta^2 c^2 = -7 \implies \beta = \frac{1}{2}\sqrt{7} \quad \text{and} \quad b = \frac{1}{2}\sqrt{7}i \\ a^2 - p^2 c^2 &= 0 \implies p = 0. \end{aligned}$$

Then, by (19) in the text,

$$y = c_1 J_0 \left(\frac{1}{2} \sqrt{7} i x^2 \right) + c_2 Y_0 \left(\frac{1}{2} \sqrt{7} i x^2 \right).$$

In terms of real functions the general solution can be written

$$y = C_1 I_0 \left(\frac{1}{2} \sqrt{7} x^2 \right) + C_2 K_0 \left(\frac{1}{2} \sqrt{7} x^2 \right).$$

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23. The differential equation has the form of (18) in the text with

$$\begin{aligned}1 - 2a &= 0 \implies a = \frac{1}{2} \\ 2c - 2 &= 0 \implies c = 1 \\ b^2 c^2 &= 1 \implies b = 1 \\ a^2 - p^2 c^2 &= 0 \implies p = \frac{1}{2}.\end{aligned}$$

Then, by (19) in the text,

$$y = x^{1/2}[c_1 J_{1/2}(x) + c_2 J_{-1/2}(x)] = x^{1/2} \left[c_1 \sqrt{\frac{2}{\pi x}} \sin x + c_2 \sqrt{\frac{2}{\pi x}} \cos x \right] = C_1 \sin x + C_2 \cos x.$$

24. Write the differential equation in the form $y'' + (4/x)y' + (1 + 2/x^2)y = 0$. This is the form of (18) in the text with

$$\begin{aligned}1 - 2a &= 4 \implies a = -\frac{3}{2} \\ 2c - 2 &= 0 \implies c = 1 \\ b^2 c^2 &= 1 \implies b = 1 \\ a^2 - p^2 c^2 &= 2 \implies p = \frac{1}{2}.\end{aligned}$$

Then, by (19), (23), and (24) in the text,

$$y = x^{-3/2}[c_1 J_{1/2}(x) + c_2 J_{-1/2}(x)] = x^{-3/2} \left[c_1 \sqrt{\frac{2}{\pi x}} \sin x + c_2 \sqrt{\frac{2}{\pi x}} \cos x \right] = C_1 \frac{1}{x^2} \sin x + C_2 \frac{1}{x^2} \cos x.$$

25. Write the differential equation in the form $y'' + (2/x)y' + (\frac{1}{16}x^2 - 3/4x^2)y = 0$. This is the form of (18) in the text with

$$\begin{aligned}1 - 2a &= 2 \implies a = -\frac{1}{2} \\ 2c - 2 &= 2 \implies c = 2 \\ b^2 c^2 &= \frac{1}{16} \implies b = \frac{1}{8} \\ a^2 - p^2 c^2 &= -\frac{3}{4} \implies p = \frac{1}{2}.\end{aligned}$$

Then, by (19) in the text,

$$\begin{aligned}y &= x^{-1/2} \left[c_1 J_{1/2} \left(\frac{1}{8}x^2 \right) + c_2 J_{-1/2} \left(\frac{1}{8}x^2 \right) \right] \\ &= x^{-1/2} \left[c_1 \sqrt{\frac{16}{\pi x^2}} \sin \left(\frac{1}{8}x^2 \right) + c_2 \sqrt{\frac{16}{\pi x^2}} \cos \left(\frac{1}{8}x^2 \right) \right] \\ &= C_1 x^{-3/2} \sin \left(\frac{1}{8}x^2 \right) + C_2 x^{-3/2} \cos \left(\frac{1}{8}x^2 \right).\end{aligned}$$

26. Write the differential equation in the form $y'' - (1/x)y' + (4 + 3/4x^2)y = 0$. This is the form of (18) in the text with

$$\begin{aligned}1 - 2a &= -1 \implies a = 1 \\ 2c - 2 &= 0 \implies c = 1 \\ b^2 c^2 &= 4 \implies b = 2 \\ a^2 - p^2 c^2 &= \frac{3}{4} \implies p = \frac{1}{2}.\end{aligned}$$

Then, by (19) in the text,

$$\begin{aligned} y &= x[c_1 J_{1/2}(2x) + c_2 J_{-1/2}(2x)] \\ &= x \left[c_1 \sqrt{\frac{2}{\pi 2x}} \sin 2x + c_2 \sqrt{\frac{2}{\pi 2x}} \cos 2x \right] \\ &= C_1 x^{1/2} \sin 2x + C_2 x^{1/2} \cos 2x. \end{aligned}$$

27. (a) The recurrence relation follows from

$$\begin{aligned} -\nu J_\nu(x) + x J_{\nu-1}(x) &= - \sum_{n=0}^{\infty} \frac{(-1)^n \nu}{n! \Gamma(1 + \nu + n)} \left(\frac{x}{2}\right)^{2n+\nu} + x \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(\nu + n)} \left(\frac{x}{2}\right)^{2n+\nu-1} \\ &= - \sum_{n=0}^{\infty} \frac{(-1)^n \nu}{n! \Gamma(1 + \nu + n)} \left(\frac{x}{2}\right)^{2n+\nu} + \sum_{n=0}^{\infty} \frac{(-1)^n (\nu + n)}{n! \Gamma(1 + \nu + n)} \cdot 2 \left(\frac{x}{2}\right) \left(\frac{x}{2}\right)^{2n+\nu-1} \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n (2n + \nu)}{n! \Gamma(1 + \nu + n)} \left(\frac{x}{2}\right)^{2n+\nu} = x J'_\nu(x). \end{aligned}$$

(b) The formula in part (a) is a linear first-order differential equation in $J_\nu(x)$. An integrating factor for this equation is x^ν , so

$$\frac{d}{dx}[x^\nu J_\nu(x)] = x^\nu J_{\nu-1}(x).$$

28. Subtracting the formula in part (a) of Problem 27 from the formula in Example 5 we obtain

$$0 = 2\nu J_\nu(x) - x J_{\nu+1}(x) - x J_{\nu-1}(x) \quad \text{or} \quad 2\nu J_\nu(x) = x J_{\nu+1}(x) + x J_{\nu-1}(x).$$

29. Letting $\nu = 1$ in (21) in the text we have

$$x J_0(x) = \frac{d}{dx}[x J_1(x)] \quad \text{so} \quad \int_0^x r J_0(r) dr = r J_1(r) \Big|_{r=0}^{r=x} = x J_1(x).$$

30. From (20) we obtain $J'_0(x) = -J_1(x)$, and from (21) we obtain $J'_0(x) = J_{-1}(x)$. Thus $J'_0(x) = J_{-1}(x) = -J_1(x)$.

31. Since $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ and

$$\Gamma\left(1 - \frac{1}{2} + n\right) = \frac{(2n-1)!}{(n-1)! 2^{2n-1}} \sqrt{\pi} \quad n = 1, 2, 3, \dots,$$

we obtain

$$\begin{aligned} J_{-1/2}(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n}{n! \Gamma(1 - \frac{1}{2} + n)} \left(\frac{x}{2}\right)^{2n-1/2} = \frac{1}{\Gamma(\frac{1}{2})} \left(\frac{x}{2}\right)^{-1/2} + \sum_{n=1}^{\infty} \frac{(-1)^n (n-1)! 2^{2n-1} x^{2n-1/2}}{n! (2n-1)! 2^{2n-1/2} \sqrt{\pi}} \\ &= \frac{1}{\sqrt{\pi}} \sqrt{\frac{2}{x}} + \sum_{n=1}^{\infty} \frac{(-1)^n 2^{1/2} x^{-1/2}}{2n(2n-1)! \sqrt{\pi}} x^{2n} = \sqrt{\frac{2}{\pi x}} + \sqrt{\frac{2}{\pi x}} \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = \sqrt{\frac{2}{\pi x}} \cos x. \end{aligned}$$

32. (a) By Problem 28, with $\nu = 1/2$, we obtain $J_{1/2}(x) = x J_{3/2}(x) + x J_{-1/2}(x)$ so that

$$J_{3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right);$$

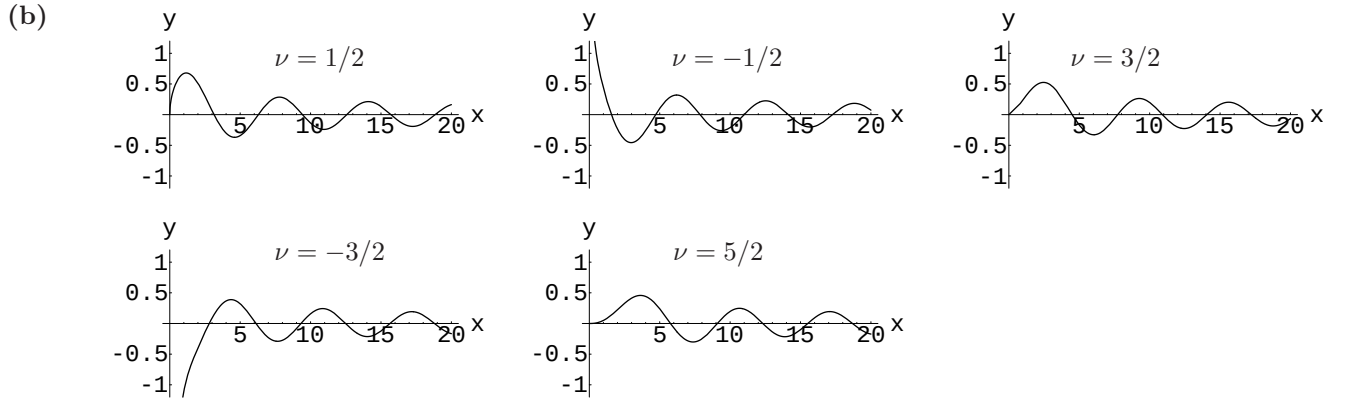
with $\nu = -1/2$ we obtain $-J_{-1/2}(x) = x J_{1/2}(x) + x J_{-3/2}(x)$ so that

$$J_{-3/2}(x) = -\sqrt{\frac{2}{\pi x}} \left(\frac{\cos x}{x} + \sin x \right);$$

and with $\nu = 3/2$ we obtain $3J_{3/2}(x) = x J_{5/2}(x) + x J_{1/2}(x)$ so that

$$J_{5/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{3 \sin x}{x^2} - \frac{3 \cos x}{x} - \sin x \right).$$

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33. Letting

$$s = \frac{2}{\alpha} \sqrt{\frac{k}{m}} e^{-\alpha t/2},$$

we have

$$\frac{dx}{dt} = \frac{dx}{ds} \frac{ds}{dt} = \frac{dx}{ds} \left[\frac{2}{\alpha} \sqrt{\frac{k}{m}} \left(-\frac{\alpha}{2} \right) e^{-\alpha t/2} \right] = \frac{dx}{ds} \left(-\sqrt{\frac{k}{m}} e^{-\alpha t/2} \right)$$

and

$$\begin{aligned} \frac{d^2x}{dt^2} &= \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{dx}{ds} \left(\frac{\alpha}{2} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) + \frac{d}{dt} \left(\frac{dx}{ds} \right) \left(-\sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) \\ &= \frac{dx}{ds} \left(\frac{\alpha}{2} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) + \frac{d^2x}{ds^2} \frac{ds}{dt} \left(-\sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) \\ &= \frac{dx}{ds} \left(\frac{\alpha}{2} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \right) + \frac{d^2x}{ds^2} \left(\frac{k}{m} e^{-\alpha t} \right). \end{aligned}$$

Then

$$m \frac{d^2x}{dt^2} + k e^{-\alpha t} x = k e^{-\alpha t} \frac{d^2x}{ds^2} + \frac{m\alpha}{2} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \frac{dx}{ds} + k e^{-\alpha t} x = 0.$$

Multiplying by $2^2/\alpha^2 m$ we have

$$\frac{2^2}{\alpha^2} \frac{k}{m} e^{-\alpha t} \frac{d^2x}{ds^2} + \frac{2}{\alpha} \sqrt{\frac{k}{m}} e^{-\alpha t/2} \frac{dx}{ds} + \frac{2^2}{\alpha^2} \frac{k}{m} e^{-\alpha t} x = 0$$

or, since $s = (2/\alpha) \sqrt{k/m} e^{-\alpha t/2}$,

$$s^2 \frac{d^2x}{ds^2} + s \frac{dx}{ds} + s^2 x = 0.$$

34. Differentiating $y = x^{1/2} w \left(\frac{2}{3} \alpha x^{3/2} \right)$ with respect to $\frac{2}{3} \alpha x^{3/2}$ we obtain

$$y' = x^{1/2} w' \left(\frac{2}{3} \alpha x^{3/2} \right) \alpha x^{1/2} + \frac{1}{2} x^{-1/2} w \left(\frac{2}{3} \alpha x^{3/2} \right)$$

and

$$\begin{aligned} y'' &= \alpha x w'' \left(\frac{2}{3} \alpha x^{3/2} \right) \alpha x^{1/2} + \alpha w' \left(\frac{2}{3} \alpha x^{3/2} \right) \\ &\quad + \frac{1}{2} \alpha w' \left(\frac{2}{3} \alpha x^{3/2} \right) - \frac{1}{4} x^{-3/2} w \left(\frac{2}{3} \alpha x^{3/2} \right). \end{aligned}$$

Then, after combining terms and simplifying, we have

$$y'' + \alpha^2 x y = \alpha \left[\alpha x^{3/2} w'' + \frac{3}{2} w' + \left(\alpha x^{3/2} - \frac{1}{4 \alpha x^{3/2}} \right) w \right] = 0.$$

Letting $t = \frac{2}{3}\alpha x^{3/2}$ or $\alpha x^{3/2} = \frac{3}{2}t$ this differential equation becomes

$$\frac{3}{2} \frac{\alpha}{t} \left[t^2 w''(t) + t w'(t) + \left(t^2 - \frac{1}{9} \right) w(t) \right] = 0, \quad t > 0.$$

- 35. (a)** By Problem 34, a solution of Airy's equation is $y = x^{1/2} w(\frac{2}{3}\alpha x^{3/2})$, where

$$w(t) = c_1 J_{1/3}(t) + c_2 J_{-1/3}(t)$$

is a solution of Bessel's equation of order $\frac{1}{3}$. Thus, the general solution of Airy's equation for $x > 0$ is

$$y = x^{1/2} w\left(\frac{2}{3}\alpha x^{3/2}\right) = c_1 x^{1/2} J_{1/3}\left(\frac{2}{3}\alpha x^{3/2}\right) + c_2 x^{1/2} J_{-1/3}\left(\frac{2}{3}\alpha x^{3/2}\right).$$

- (b)** Airy's equation, $y'' + \alpha^2 xy = 0$, has the form of (18) in the text with

$$\begin{aligned} 1 - 2a &= 0 \implies a = \frac{1}{2} \\ 2c - 2 &= 1 \implies c = \frac{3}{2} \\ b^2 c^2 &= \alpha^2 \implies b = \frac{2}{3}\alpha \\ a^2 - p^2 c^2 &= 0 \implies p = \frac{1}{3}. \end{aligned}$$

Then, by (19) in the text,

$$y = x^{1/2} \left[c_1 J_{1/3}\left(\frac{2}{3}\alpha x^{3/2}\right) + c_2 J_{-1/3}\left(\frac{2}{3}\alpha x^{3/2}\right) \right].$$

- 36.** The general solution of the differential equation is

$$y(x) = c_1 J_0(\alpha x) + c_2 Y_0(\alpha x).$$

In order to satisfy the conditions that $\lim_{x \rightarrow 0^+} y(x)$ and $\lim_{x \rightarrow 0^+} y'(x)$ are finite we are forced to define $c_2 = 0$. Thus, $y(x) = c_1 J_0(\alpha x)$. The second boundary condition, $y(2) = 0$, implies $c_1 = 0$ or $J_0(2\alpha) = 0$. In order to have a nontrivial solution we require that $J_0(2\alpha) = 0$. From Table 5.1, the first three positive zeros of J_0 are found to be

$$2\alpha_1 = 2.4048, \quad 2\alpha_2 = 5.5201, \quad 2\alpha_3 = 8.6537$$

and so $\alpha_1 = 1.2024$, $\alpha_2 = 2.7601$, $\alpha_3 = 4.3269$. The eigenfunctions corresponding to the eigenvalues $\lambda_1 = \alpha_1^2$, $\lambda_2 = \alpha_2^2$, $\lambda_3 = \alpha_3^2$ are $J_0(1.2024x)$, $J_0(2.7601x)$, and $J_0(4.3269x)$.

- 37. (a)** The differential equation $y'' + (\lambda/x)y = 0$ has the form of (18) in the text with

$$\begin{aligned} 1 - 2a &= 0 \implies a = \frac{1}{2} \\ 2c - 2 &= -1 \implies c = \frac{1}{2} \\ b^2 c^2 &= \lambda \implies b = 2\sqrt{\lambda} \\ a^2 - p^2 c^2 &= 0 \implies p = 1. \end{aligned}$$

Then, by (19) in the text,

$$y = x^{1/2} [c_1 J_1(2\sqrt{\lambda x}) + c_2 Y_1(2\sqrt{\lambda x})].$$

- (b)** We first note that $y = J_1(t)$ is a solution of Bessel's equation, $t^2 y'' + t y' + (t^2 - 1)y = 0$, with $\nu = 1$. That is,

$$t^2 J_1''(t) + t J_1'(t) + (t^2 - 1)J_1(t) = 0,$$

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or, letting $t = 2\sqrt{x}$,

$$4xJ_1''(2\sqrt{x}) + 2\sqrt{x}J_1'(2\sqrt{x}) + (4x - 1)J_1(2\sqrt{x}) = 0.$$

Now, if $y = \sqrt{x}J_1(2\sqrt{x})$, we have

$$y' = \sqrt{x}J_1'(2\sqrt{x})\frac{1}{\sqrt{x}} + \frac{1}{2\sqrt{x}}J_1(2\sqrt{x}) = J_1'(2\sqrt{x}) + \frac{1}{2}x^{-1/2}J_1(2\sqrt{x})$$

and

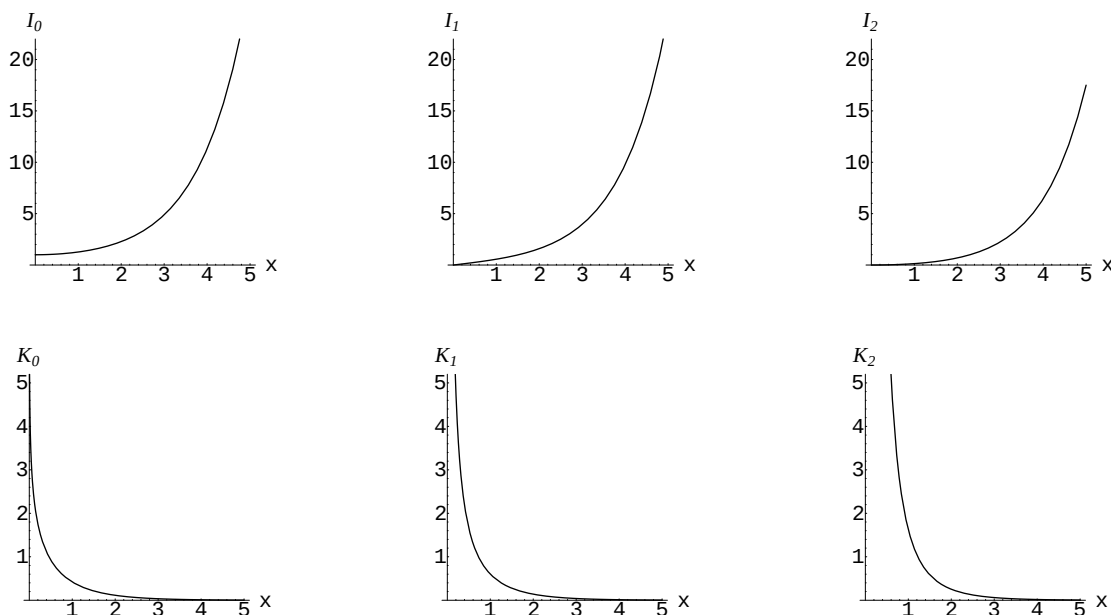
$$y'' = x^{-1/2}J_1''(2\sqrt{x}) + \frac{1}{2x}J_1'(2\sqrt{x}) - \frac{1}{4}x^{-3/2}J_1(2\sqrt{x}).$$

Then

$$\begin{aligned} xy'' + y &= \sqrt{x}J_1''2\sqrt{x} + \frac{1}{2}J_1'(2\sqrt{x}) - \frac{1}{4}x^{-1/2}J_1(2\sqrt{x}) + \sqrt{x}J_1(2\sqrt{x}) \\ &= \frac{1}{4\sqrt{x}}[4xJ_1''(2\sqrt{x}) + 2\sqrt{x}J_1'(2\sqrt{x}) - J_1(2\sqrt{x}) + 4xJ_1(2\sqrt{x})] \\ &= 0, \end{aligned}$$

and $y = \sqrt{x}J_1(2\sqrt{x})$ is a solution of Airy's differential equation.

- 38.** We see from the graphs below that the graphs of the modified Bessel functions are not oscillatory, while those of the Bessel functions, shown in Figures 5.3 and 5.4 in the text, are oscillatory.



- 39. (a)** We identify $m = 4$, $k = 1$, and $\alpha = 0.1$. Then

$$x(t) = c_1J_0(10e^{-0.05t}) + c_2Y_0(10e^{-0.05t})$$

and

$$x'(t) = -0.5c_1J_0'(10e^{-0.05t}) - 0.5c_2Y_0'(10e^{-0.05t}).$$

Now $x(0) = 1$ and $x'(0) = -1/2$ imply

$$c_1J_0(10) + c_2Y_0(10) = 1$$

$$c_1J_0'(10) + c_2Y_0'(10) = 1.$$

Using Cramer's rule we obtain

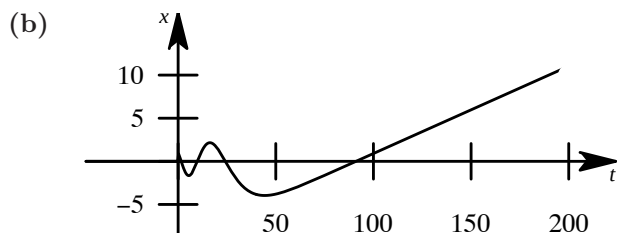
$$c_1 = \frac{Y_0'(10) - Y_0(10)}{J_0(10)Y_0'(10) - J_0'(10)Y_0(10)}$$

and

$$c_2 = \frac{J_0(10) - J_0'(10)}{J_0(10)Y_0'(10) - J_0'(10)Y_0(10)}.$$

Using $Y_0' = -Y_1$ and $J_0' = -J_1$ and Table 5.2 we find $c_1 = -4.7860$ and $c_2 = -3.1803$. Thus

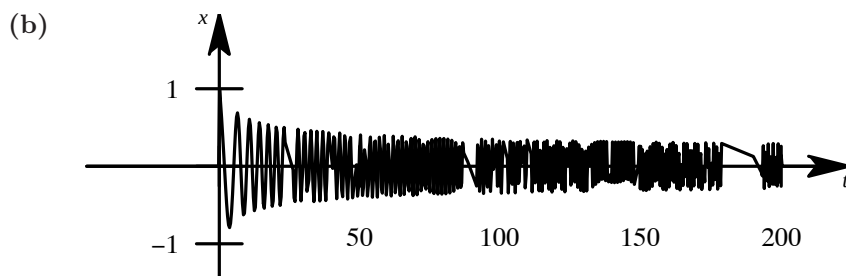
$$x(t) = -4.7860J_0(10e^{-0.05t}) - 3.1803Y_0(10e^{-0.05t}).$$



40. (a) Identifying $\alpha = \frac{1}{2}$, the general solution of $x'' + \frac{1}{4}tx = 0$ is

$$x(t) = c_1 x^{1/2} J_{1/3} \left(\frac{1}{3} x^{3/2} \right) + c_2 x^{1/2} J_{-1/3} \left(\frac{1}{3} x^{3/2} \right).$$

Solving the system $x(0.1) = 1$, $x'(0.1) = -\frac{1}{2}$ we find $c_1 = -0.809264$ and $c_2 = 0.782397$.



41. (a) Letting $t = L - x$, the boundary-value problem becomes

$$\frac{d^2\theta}{dt^2} + \alpha^2 t\theta = 0, \quad \theta'(0) = 0, \quad \theta(L) = 0,$$

where $\alpha^2 = \delta g/EI$. This is Airy's differential equation, so by Problem 35 its solution is

$$y = c_1 t^{1/2} J_{1/3} \left(\frac{2}{3} \alpha t^{3/2} \right) + c_2 t^{1/2} J_{-1/3} \left(\frac{2}{3} \alpha t^{3/2} \right) = c_1 \theta_1(t) + c_2 \theta_2(t).$$

- (b) Looking at the series forms of θ_1 and θ_2 we see that $\theta_1'(0) \neq 0$, while $\theta_2'(0) = 0$. Thus, the boundary condition $\theta'(0) = 0$ implies $c_1 = 0$, and so

$$\theta(t) = c_2 \sqrt{t} J_{-1/3} \left(\frac{2}{3} \alpha t^{3/2} \right).$$

From $\theta(L) = 0$ we have

$$c_2 \sqrt{L} J_{-1/3} \left(\frac{2}{3} \alpha L^{3/2} \right) = 0,$$

so either $c_2 = 0$, in which case $\theta(t) = 0$, or $J_{-1/3}(\frac{2}{3}\alpha L^{3/2}) = 0$. The column will just start to bend when L is the length corresponding to the smallest positive zero of $J_{-1/3}$.

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(c) Using *Mathematica*, the first positive root of $J_{-1/3}(x)$ is $x_1 \approx 1.86635$. Thus $\frac{2}{3}\alpha L^{3/2} = 1.86635$ implies

$$\begin{aligned} L &= \left(\frac{3(1.86635)}{2\alpha} \right)^{2/3} = \left[\frac{9EI}{4\delta g} (1.86635)^2 \right]^{1/3} \\ &= \left[\frac{9(2.6 \times 10^7)\pi(0.05)^4/4}{4(0.28)\pi(0.05)^2} (1.86635)^2 \right]^{1/3} \approx 76.9 \text{ in.} \end{aligned}$$

42. (a) Writing the differential equation in the form $xy'' + (PL/M)y = 0$, we identify $\lambda = PL/M$. From Problem 37 the solution of this differential equation is

$$y = c_1 \sqrt{x} J_1 \left(2\sqrt{PLx/M} \right) + c_2 \sqrt{x} Y_1 \left(2\sqrt{PLx/M} \right).$$

Now $J_1(0) = 0$, so $y(0) = 0$ implies $c_2 = 0$ and

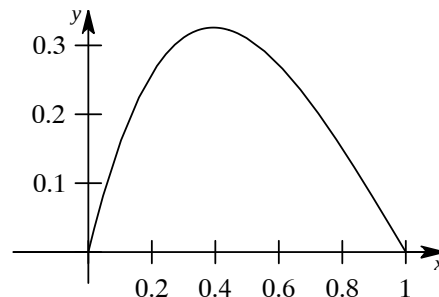
$$y = c_1 \sqrt{x} J_1 \left(2\sqrt{PLx/M} \right).$$

(b) From $y(L) = 0$ we have $y = J_1(2L\sqrt{PM}) = 0$. The first positive zero of J_1 is 3.8317 so, solving

$2L\sqrt{P_1/M} = 3.8317$, we find $P_1 = 3.6705M/L^2$. Therefore,

$$y_1(x) = c_1 \sqrt{x} J_1 \left(2\sqrt{\frac{3.6705x}{L}} \right) = c_1 \sqrt{x} J_1 \left(\frac{3.8317}{\sqrt{L}} \sqrt{x} \right).$$

(c) For $c_1 = 1$ and $L = 1$ the graph of $y_1 = \sqrt{x} J_1(3.8317\sqrt{x})$ is shown.



43. (a) Since $l' = v$, we integrate to obtain $l(t) = vt + c$. Now $l(0) = l_0$ implies $c = l_0$, so $l(t) = vt + l_0$. Using $\sin \theta \approx \theta$ in $l d^2\theta/dt^2 + 2l' d\theta/dt + g \sin \theta = 0$ gives

$$(l_0 + vt) \frac{d^2\theta}{dt^2} + 2v \frac{d\theta}{dt} + g\theta = 0.$$

(b) Dividing by v , the differential equation in part (a) becomes

$$\frac{l_0 + vt}{v} \frac{d^2\theta}{dt^2} + 2 \frac{d\theta}{dt} + \frac{g}{v} \theta = 0.$$

Letting $x = (l_0 + vt)/v = t + l_0/v$ we have $dx/dt = 1$, so

$$\frac{d\theta}{dt} = \frac{d\theta}{dx} \frac{dx}{dt} = \frac{d\theta}{dx}$$

and

$$\frac{d^2\theta}{dt^2} = \frac{d(d\theta/dt)}{dt} = \frac{d(d\theta/dx)}{dx} \frac{dx}{dt} = \frac{d^2\theta}{dx^2}.$$

Thus, the differential equation becomes

$$x \frac{d^2\theta}{dx^2} + 2 \frac{d\theta}{dx} + \frac{g}{v} \theta = 0 \quad \text{or} \quad \frac{d^2\theta}{dx^2} + \frac{2}{x} \frac{d\theta}{dx} + \frac{g}{vx} \theta = 0.$$

(c) The differential equation in part (b) has the form of (18) in the text with

$$\begin{aligned} 1 - 2a = 2 &\implies a = -\frac{1}{2} \\ 2c - 2 = -1 &\implies c = \frac{1}{2} \\ b^2 c^2 = \frac{g}{v} &\implies b = 2\sqrt{\frac{g}{v}} \\ a^2 - p^2 c^2 = 0 &\implies p = 1. \end{aligned}$$

Then, by (19) in the text,

$$\theta(x) = x^{-1/2} \left[c_1 J_1 \left(2\sqrt{\frac{g}{v}} x^{1/2} \right) + c_2 Y_1 \left(2\sqrt{\frac{g}{v}} x^{1/2} \right) \right]$$

or

$$\theta(t) = \sqrt{\frac{v}{l_0 + vt}} \left[c_1 J_1 \left(\frac{2}{v} \sqrt{g(l_0 + vt)} \right) + c_2 Y_1 \left(\frac{2}{v} \sqrt{g(l_0 + vt)} \right) \right].$$

(d) To simplify calculations, let

$$u = \frac{2}{v} \sqrt{g(l_0 + vt)} = 2\sqrt{\frac{g}{v}} x^{1/2},$$

and at $t = 0$ let $u_0 = 2\sqrt{gl_0}/v$. The general solution for $\theta(t)$ can then be written

$$\theta = C_1 u^{-1} J_1(u) + C_2 u^{-1} Y_1(u). \quad (1)$$

Before applying the initial conditions, note that

$$\frac{d\theta}{dt} = \frac{d\theta}{du} \frac{du}{dt}$$

so when $d\theta/dt = 0$ at $t = 0$ we have $d\theta/du = 0$ at $u = u_0$. Also,

$$\frac{d\theta}{du} = C_1 \frac{d}{du} [u^{-1} J_1(u)] + C_2 \frac{d}{du} [u^{-1} Y_1(u)]$$

which, in view of (20) in the text, is the same as

$$\frac{d\theta}{du} = -C_1 u^{-1} J_2(u) - C_2 u^{-1} Y_2(u). \quad (2)$$

Now at $t = 0$, or $u = u_0$, (1) and (2) give the system

$$\begin{aligned} C_1 u_0^{-1} J_1(u_0) + C_2 u_0^{-1} Y_1(u_0) &= \theta_0 \\ C_1 u_0^{-1} J_2(u_0) + C_2 u_0^{-1} Y_2(u_0) &= 0 \end{aligned}$$

whose solution is easily obtained using Cramer's rule:

$$C_1 = \frac{u_0 \theta_0 Y_2(u_0)}{J_1(u_0) Y_2(u_0) - J_2(u_0) Y_1(u_0)}, \quad C_2 = \frac{-u_0 \theta_0 J_2(u_0)}{J_1(u_0) Y_2(u_0) - J_2(u_0) Y_1(u_0)}.$$

In view of the given identity these results simplify to

$$C_1 = -\frac{\pi}{2} u_0^2 \theta_0 Y_2(u_0) \quad \text{and} \quad C_2 = \frac{\pi}{2} u_0^2 \theta_0 J_2(u_0).$$

The solution is then

$$\theta = \frac{\pi}{2} u_0^2 \theta_0 \left[-Y_2(u_0) \frac{J_1(u)}{u} + J_2(u_0) \frac{Y_1(u)}{u} \right].$$

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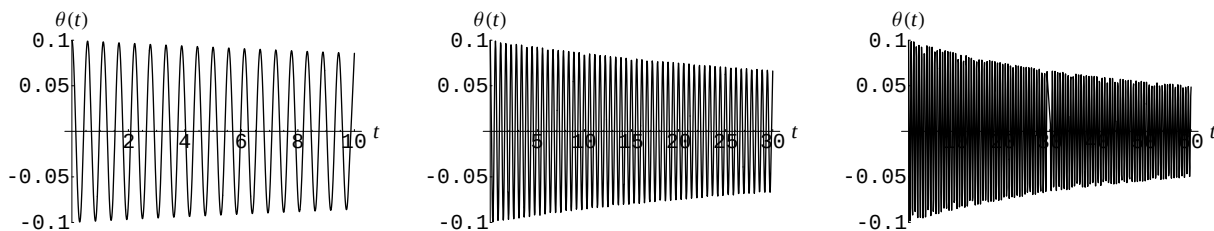
Returning to $u = (2/v)\sqrt{g(l_0 + vt)}$ and $u_0 = (2/v)\sqrt{gl_0}$, we have

$$\theta(t) = \frac{\pi\sqrt{gl_0}\theta_0}{v} \left[-Y_2\left(\frac{2}{v}\sqrt{gl_0}\right) \frac{J_1\left(\frac{2}{v}\sqrt{g(l_0 + vt)}\right)}{\sqrt{l_0 + vt}} + J_2\left(\frac{2}{v}\sqrt{gl_0}\right) \frac{Y_1\left(\frac{2}{v}\sqrt{g(l_0 + vt)}\right)}{\sqrt{l_0 + vt}} \right].$$

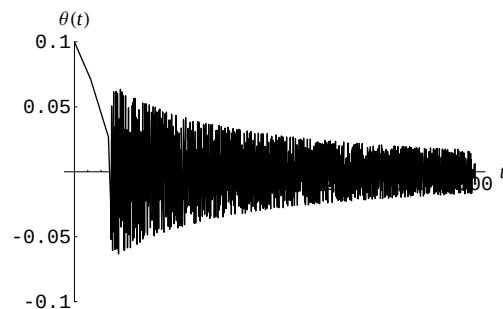
(e) When $l_0 = 1$ ft, $\theta_0 = \frac{1}{10}$ radian, and $v = \frac{1}{60}$ ft/s, the above function is

$$\theta(t) = -1.69045 \frac{J_1(480\sqrt{2}(1 + t/60))}{\sqrt{1 + t/60}} - 2.79381 \frac{Y_1(480\sqrt{2}(1 + t/60))}{\sqrt{1 + t/60}}.$$

The plots of $\theta(t)$ on $[0, 10]$, $[0, 30]$, and $[0, 60]$ are



(f) The graphs indicate that $\theta(t)$ decreases as l increases. The graph of $\theta(t)$ on $[0, 300]$ is shown.



44. (a) From (26) in the text, we have

$$P_6(x) = c_0 \left(1 - \frac{6 \cdot 7}{2!} x^2 + \frac{4 \cdot 6 \cdot 7 \cdot 9}{4!} x^4 - \frac{2 \cdot 4 \cdot 6 \cdot 7 \cdot 9 \cdot 11}{6!} x^6 \right),$$

where

$$c_0 = (-1)^3 \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} = -\frac{5}{16}.$$

Thus,

$$P_6(x) = -\frac{5}{16} \left(1 - 21x^2 + 63x^4 - \frac{231}{5}x^6 \right) = \frac{1}{16}(231x^6 - 315x^4 + 105x^2 - 5).$$

Also, from (26) in the text we have

$$P_7(x) = c_1 \left(x - \frac{6 \cdot 9}{3!} x^3 + \frac{4 \cdot 6 \cdot 9 \cdot 11}{5!} x^5 - \frac{2 \cdot 4 \cdot 6 \cdot 9 \cdot 11 \cdot 13}{7!} x^7 \right)$$

where

$$c_1 = (-1)^3 \frac{1 \cdot 3 \cdot 5 \cdot 7}{2 \cdot 4 \cdot 6} = -\frac{35}{16}.$$

Thus

$$P_7(x) = -\frac{35}{16} \left(x - 9x^3 + \frac{99}{5}x^5 - \frac{429}{35}x^7 \right) = \frac{1}{16}(429x^7 - 693x^5 + 315x^3 - 35x).$$

(b) $P_6(x)$ satisfies $(1 - x^2)y'' - 2xy' + 42y = 0$ and $P_7(x)$ satisfies $(1 - x^2)y'' - 2xy' + 56y = 0$.

45. The recurrence relation can be written

$$P_{k+1}(x) = \frac{2k+1}{k+1} x P_k(x) - \frac{k}{k+1} P_{k-1}(x), \quad k = 2, 3, 4, \dots$$

$$k = 1: P_2(x) = \frac{3}{2}x^2 - \frac{1}{2}$$

$$k = 2: P_3(x) = \frac{5}{3}x \left(\frac{3}{2}x^2 - \frac{1}{2} \right) - \frac{2}{3}x = \frac{5}{2}x^3 - \frac{3}{2}x$$

$$k = 3: P_4(x) = \frac{7}{4}x \left(\frac{5}{2}x^3 - \frac{3}{2}x \right) - \frac{3}{4} \left(\frac{3}{2}x^2 - \frac{1}{2} \right) = \frac{35}{8}x^4 - \frac{30}{8}x^2 + \frac{3}{8}$$

$$k = 4: P_5(x) = \frac{9}{5}x \left(\frac{35}{8}x^4 - \frac{30}{8}x^2 + \frac{3}{8} \right) - \frac{4}{5} \left(\frac{5}{2}x^3 - \frac{3}{2}x \right) = \frac{63}{8}x^5 - \frac{35}{4}x^3 + \frac{15}{8}x$$

$$k = 5: P_6(x) = \frac{11}{6}x \left(\frac{63}{8}x^5 - \frac{35}{4}x^3 + \frac{15}{8}x \right) - \frac{5}{6} \left(\frac{35}{8}x^4 - \frac{30}{8}x^2 + \frac{3}{8} \right) = \frac{231}{16}x^6 - \frac{315}{16}x^4 + \frac{105}{16}x^2 - \frac{5}{16}$$

$$\begin{aligned} k = 6: P_7(x) &= \frac{13}{7}x \left(\frac{231}{16}x^6 - \frac{315}{16}x^4 + \frac{105}{16}x^2 - \frac{5}{16} \right) - \frac{6}{7} \left(\frac{63}{8}x^5 - \frac{35}{4}x^3 + \frac{15}{8}x \right) \\ &= \frac{429}{16}x^7 - \frac{693}{16}x^5 + \frac{315}{16}x^3 - \frac{35}{16}x \end{aligned}$$

46. If $x = \cos \theta$ then

$$\frac{dy}{d\theta} = -\sin \theta \frac{dy}{dx},$$

$$\frac{d^2y}{d\theta^2} = \sin^2 \theta \frac{d^2y}{dx^2} - \cos \theta \frac{dy}{dx},$$

and

$$\sin \theta \frac{d^2y}{d\theta^2} + \cos \theta \frac{dy}{d\theta} + n(n+1)(\sin \theta)y = \sin \theta \left[(1 - \cos^2 \theta) \frac{d^2y}{dx^2} - 2 \cos \theta \frac{dy}{dx} + n(n+1)y \right] = 0.$$

That is,

$$(1 - x^2) \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0.$$

47. The only solutions bounded on $[-1, 1]$ are $y = cP_n(x)$, c a constant and $n = 0, 1, 2, \dots$. By (iv) of the properties of the Legendre polynomials, $y(0) = 0$ or $P_n(0) = 0$ implies n must be odd. Thus the first three positive eigenvalues correspond to $n = 1, 3$, and 5 or $\lambda_1 = 1 \cdot 2$, $\lambda_2 = 3 \cdot 4 = 12$, and $\lambda_3 = 5 \cdot 6 = 30$. We can take the eigenfunctions to be $y_1 = P_1(x)$, $y_2 = P_3(x)$, and $y_3 = P_5(x)$.

48. Using a CAS we find

$$P_1(x) = \frac{1}{2} \frac{d}{dx} (x^2 - 1)^1 = x$$

$$P_2(x) = \frac{1}{2^2 2!} \frac{d^2}{dx^2} (x^2 - 1)^2 = \frac{1}{2} (3x^2 - 1)$$

$$P_3(x) = \frac{1}{2^3 3!} \frac{d^3}{dx^3} (x^2 - 1)^3 = \frac{1}{2} (5x^3 - 3x)$$

$$P_4(x) = \frac{1}{2^4 4!} \frac{d^4}{dx^4} (x^2 - 1)^4 = \frac{1}{8} (35x^4 - 30x^2 + 3)$$

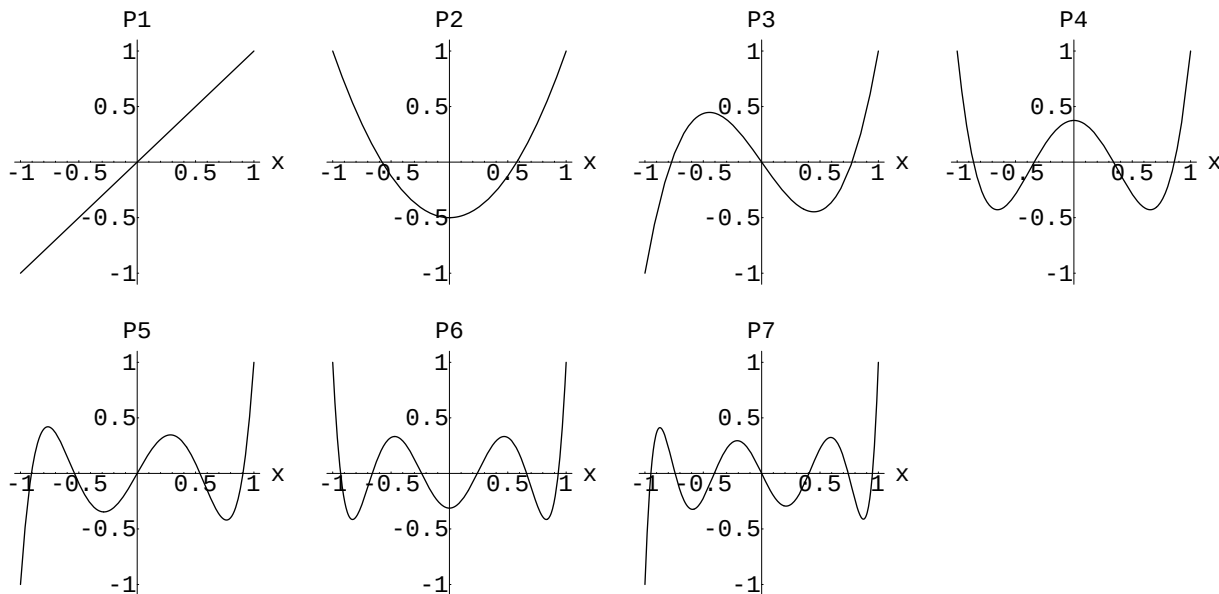
$$P_5(x) = \frac{1}{2^5 5!} \frac{d^5}{dx^5} (x^2 - 1)^5 = \frac{1}{8} (63x^5 - 70x^3 + 15x)$$

$$P_6(x) = \frac{1}{2^6 6!} \frac{d^6}{dx^6} (x^2 - 1)^6 = \frac{1}{16} (231x^6 - 315x^4 + 105x^2 - 5)$$

$$P_7(x) = \frac{1}{2^7 7!} \frac{d^7}{dx^7} (x^2 - 1)^7 = \frac{1}{16} (429x^7 - 693x^5 + 315x^3 - 35x)$$

5.3 Special Functions

49.



50. Zeros of Legendre polynomials for $n \geq 1$ are

$$P_1(x) : 0$$

$$P_2(x) : \pm 0.57735$$

$$P_3(x) : 0, \pm 0.77460$$

$$P_4(x) : \pm 0.33998, \pm 0.86115$$

$$P_5(x) : 0, \pm 0.53847, \pm 0.90618$$

$$P_6(x) : \pm 0.23862, \pm 0.66121, \pm 0.93247$$

$$P_7(x) : 0, \pm 0.40585, \pm 0.74153, \pm 0.94911$$

$$P_{10}(x) : \pm 0.14887, \pm 0.43340, \pm 0.67941, \pm 0.86506, \pm 0.097391$$

The zeros of any Legendre polynomial are in the interval $(-1, 1)$ and are symmetric with respect to 0.

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1. False; $J_1(x)$ and $J_{-1}(x)$ are not linearly independent when ν is a positive integer. (In this case $\nu = 1$). The general solution of $x^2 y'' + xy' + (x^2 - 1)y = 0$ is $y = c_1 J_1(x) + c_2 Y_1(x)$.
2. False; $y = x$ is a solution that is analytic at $x = 0$.
3. $x = -1$ is the nearest singular point to the ordinary point $x = 0$. Theorem 5.1 guarantees the existence of two power series solutions $y = \sum_{n=1}^{\infty} c_n x^n$ of the differential equation that converge at least for $-1 < x < 1$. Since $-\frac{1}{2} \leq x \leq \frac{1}{2}$ is properly contained in $-1 < x < 1$, both power series must converge for all points contained in $-\frac{1}{2} \leq x \leq \frac{1}{2}$.

4. The easiest way to solve the system

$$2c_2 + 2c_1 + c_0 = 0$$

$$6c_3 + 4c_2 + c_1 = 0$$

$$12c_4 + 6c_3 - \frac{1}{3}c_1 + c_2 = 0$$

$$20c_5 + 8c_4 - \frac{2}{3}c_2 + c_3 = 0$$

is to choose, in turn, $c_0 \neq 0$, $c_1 = 0$ and $c_0 = 0$, $c_1 \neq 0$. Assuming that $c_0 \neq 0$, $c_1 = 0$, we have

$$c_2 = -\frac{1}{2}c_0$$

$$c_3 = -\frac{2}{3}c_2 = \frac{1}{3}c_0$$

$$c_4 = -\frac{1}{2}c_3 - \frac{1}{12}c_2 = -\frac{1}{8}c_0$$

$$c_5 = -\frac{2}{5}c_4 + \frac{1}{30}c_2 - \frac{1}{20}c_3 = \frac{1}{60}c_0;$$

whereas the assumption that $c_0 = 0$, $c_1 \neq 0$ implies

$$c_2 = -c_1$$

$$c_3 = -\frac{2}{3}c_2 - \frac{1}{6}c_1 = \frac{1}{2}c_1$$

$$c_4 = -\frac{1}{2}c_3 + \frac{1}{36}c_1 - \frac{1}{12}c_2 = -\frac{5}{36}c_1$$

$$c_5 = -\frac{2}{5}c_4 + \frac{1}{30}c_2 - \frac{1}{20}c_3 = -\frac{1}{360}c_1.$$

five terms of two power series solutions are then

$$y_1(x) = c_0 \left[1 - \frac{1}{2}x^2 + \frac{1}{3}x^3 - \frac{1}{8}x^4 + \frac{1}{60}x^5 + \cdots \right]$$

and

$$y_2(x) = c_1 \left[x - x^2 + \frac{1}{2}x^3 - \frac{5}{36}x^4 - \frac{1}{360}x^5 + \cdots \right].$$

5. The interval of convergence is centered at 4. Since the series converges at -2 , it converges at least on the interval $[-2, 10)$. Since it diverges at 13 , it converges at most on the interval $[-5, 13)$. Thus, at -7 it does not converge, at 0 and 7 it does converge, and at 10 and 11 it might converge.
6. We have

$$f(x) = \frac{\sin x}{\cos x} = \frac{x - \frac{x^3}{6} + \frac{x^5}{120} - \cdots}{1 - \frac{x^2}{2} + \frac{x^4}{24} - \cdots} = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots.$$

7. The differential equation $(x^3 - x^2)y'' + y' + y = 0$ has a regular singular point at $x = 1$ and an irregular singular point at $x = 0$.
8. The differential equation $(x - 1)(x + 3)y'' + y = 0$ has regular singular points at $x = 1$ and $x = -3$.
9. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation we obtain

$$2xy'' + y' + y = (2r^2 - r)c_0 x^{r-1} + \sum_{k=1}^{\infty} [2(k+r)(k+r-1)c_k + (k+r)c_k + c_{k-1}]x^{k+r-1} = 0$$

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which implies

$$2r^2 - r = r(2r - 1) = 0$$

and

$$(k + r)(2k + 2r - 1)c_k + c_{k-1} = 0.$$

The indicial roots are $r = 0$ and $r = 1/2$. For $r = 0$ the recurrence relation is

$$c_k = -\frac{c_{k-1}}{k(2k-1)}, \quad k = 1, 2, 3, \dots,$$

so

$$c_1 = -c_0, \quad c_2 = \frac{1}{6}c_0, \quad c_3 = -\frac{1}{90}c_0.$$

For $r = 1/2$ the recurrence relation is

$$c_k = -\frac{c_{k-1}}{k(2k+1)}, \quad k = 1, 2, 3, \dots,$$

so

$$c_1 = -\frac{1}{3}c_0, \quad c_2 = \frac{1}{30}c_0, \quad c_3 = -\frac{1}{630}c_0.$$

Two linearly independent solutions are

$$y_1 = 1 - x + \frac{1}{6}x^2 - \frac{1}{90}x^3 + \dots$$

and

$$y_2 = x^{1/2} \left(1 - \frac{1}{3}x + \frac{1}{30}x^2 - \frac{1}{630}x^3 + \dots \right).$$

10. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' - xy' - y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} - \underbrace{\sum_{n=1}^{\infty} n c_n x^n}_{k=n} - \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k - \sum_{k=1}^{\infty} k c_k x^k - \sum_{k=0}^{\infty} c_k x^k \\ &= 2c_2 - c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - (k+1)c_k] x^k = 0. \end{aligned}$$

Thus

$$2c_2 - c_0 = 0$$

$$(k+2)(k+1)c_{k+2} - (k+1)c_k = 0$$

and

$$c_2 = \frac{1}{2}c_0$$

$$c_{k+2} = \frac{1}{k+2} c_k, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = \frac{1}{2}$$

$$c_3 = c_5 = c_7 = \dots = 0$$

$$c_4 = \frac{1}{8}$$

$$c_6 = \frac{1}{48}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned} c_2 &= c_4 = c_6 = \cdots = 0 \\ c_3 &= \frac{1}{3} \\ c_5 &= \frac{1}{15} \\ c_7 &= \frac{1}{105} \end{aligned}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{1}{2}x^2 + \frac{1}{8}x^4 + \frac{1}{48}x^6 + \cdots$$

and

$$y_2 = x + \frac{1}{3}x^3 + \frac{1}{15}x^5 + \frac{1}{105}x^7 + \cdots$$

11. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we obtain

$$(x-1)y'' + 3y = (-2c_2 + 3c_0) + \sum_{k=1}^{\infty} [(k+1)kc_{k+1} - (k+2)(k+1)c_{k+2} + 3c_k]x^k = 0$$

which implies $c_2 = 3c_0/2$ and

$$c_{k+2} = \frac{(k+1)kc_{k+1} + 3c_k}{(k+2)(k+1)}, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = \frac{3}{2}, \quad c_3 = \frac{1}{2}, \quad c_4 = \frac{5}{8}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = 0, \quad c_3 = \frac{1}{2}, \quad c_4 = \frac{1}{4}$$

and so on. Thus, two solutions are

$$y_1 = 1 + \frac{3}{2}x^2 + \frac{1}{2}x^3 + \frac{5}{8}x^4 + \cdots$$

and

$$y_2 = x + \frac{1}{2}x^3 + \frac{1}{4}x^4 + \cdots$$

12. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we obtain

$$y'' - x^2y' + xy = 2c_2 + (6c_3 + c_0)x + \sum_{k=1}^{\infty} [(k+3)(k+2)c_{k+3} - (k-1)c_k]x^{k+1} = 0$$

which implies $c_2 = 0$, $c_3 = -c_0/6$, and

$$c_{k+3} = \frac{k-1}{(k+3)(k+2)}c_k, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned} c_3 &= -\frac{1}{6} \\ c_4 &= c_7 = c_{10} = \cdots = 0 \\ c_5 &= c_8 = c_{11} = \cdots = 0 \\ c_6 &= -\frac{1}{90} \end{aligned}$$

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and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_3 = c_6 = c_9 = \cdots = 0$$

$$c_4 = c_7 = c_{10} = \cdots = 0$$

$$c_5 = c_8 = c_{11} = \cdots = 0$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{6}x^3 - \frac{1}{90}x^6 - \cdots \quad \text{and} \quad y_2 = x.$$

13. Substituting $y = \sum_{n=0}^{\infty} c_n x^{n+r}$ into the differential equation, we obtain

$$xy'' - (x+2)y' + 2y = (r^2 - 3r)c_0 x^{r-1} + \sum_{k=1}^{\infty} [(k+r)(k+r-3)c_k - (k+r-3)c_{k-1}]x^{k+r-1} = 0,$$

which implies

$$r^2 - 3r = r(r-3) = 0$$

and

$$(k+r)(k+r-3)c_k - (k+r-3)c_{k-1} = 0.$$

The indicial roots are $r_1 = 3$ and $r_2 = 0$. For $r_2 = 0$ the recurrence relation is

$$k(k-3)c_k - (k-3)c_{k-1} = 0, \quad k = 1, 2, 3, \dots$$

Then

$$c_1 - c_0 = 0$$

$$2c_2 - c_1 = 0$$

$$0c_3 - 0c_2 = 0 \implies c_3 \text{ is arbitrary}$$

and

$$c_k = \frac{1}{k}c_{k-1}, \quad k = 4, 5, 6, \dots$$

Taking $c_0 \neq 0$ and $c_3 = 0$ we obtain

$$c_1 = c_0$$

$$c_2 = \frac{1}{2}c_0$$

$$c_3 = c_4 = c_5 = \cdots = 0.$$

Taking $c_0 = 0$ and $c_3 \neq 0$ we obtain

$$c_0 = c_1 = c_2 = 0$$

$$c_4 = \frac{1}{4}c_3 = \frac{6}{4!}c_3$$

$$c_5 = \frac{1}{5 \cdot 4}c_3 = \frac{6}{5!}c_3$$

$$c_6 = \frac{1}{6 \cdot 5 \cdot 4}c_3 = \frac{6}{6!}c_3,$$

and so on. In this case we obtain the two solutions

$$y_1 = 1 + x + \frac{1}{2}x^2$$

and

$$y_2 = x^3 + \frac{6}{4!}x^4 + \frac{6}{5!}x^5 + \frac{6}{6!}x^6 + \cdots = 6e^x - 6\left(1 + x + \frac{1}{2}x^2\right).$$

14. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 (\cos x)y'' + y &= \left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 - \frac{1}{720}x^6 + \cdots\right) (2c_2 + 6c_3x + 12c_4x^2 + 20c_5x^3 + 30c_6x^4 + \cdots) + \sum_{n=0}^{\infty} c_n x^n \\
 &= \left[2c_2 + 6c_3x + (12c_4 - c_2)x^2 + (20c_5 - 3c_3)x^3 + \left(30c_6 - 6c_4 + \frac{1}{12}c_2\right)x^4 + \cdots\right] \\
 &\quad + [c_0 + c_1x + c_2x^2 + c_3x^3 + c_4x^4 + \cdots] \\
 &= (c_0 + 2c_2) + (c_1 + 6c_3)x + 12c_4x^2 + (20c_5 - 2c_3)x^3 + \left(30c_6 - 5c_4 + \frac{1}{12}c_2\right)x^4 + \cdots \\
 &= 0.
 \end{aligned}$$

Thus

$$\begin{aligned}
 c_0 + 2c_2 &= 0 \\
 c_1 + 6c_3 &= 0 \\
 12c_4 &= 0 \\
 20c_5 - 2c_3 &= 0 \\
 30c_6 - 5c_4 + \frac{1}{12}c_2 &= 0
 \end{aligned}$$

and

$$\begin{aligned}
 c_2 &= -\frac{1}{2}c_0 \\
 c_3 &= -\frac{1}{6}c_1 \\
 c_4 &= 0 \\
 c_5 &= \frac{1}{10}c_3 \\
 c_6 &= \frac{1}{6}c_4 - \frac{1}{360}c_2.
 \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = -\frac{1}{2}, \quad c_3 = 0, \quad c_4 = 0, \quad c_5 = 0, \quad c_6 = \frac{1}{720}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we find

$$c_2 = 0, \quad c_3 = -\frac{1}{6}, \quad c_4 = 0, \quad c_5 = -\frac{1}{60}, \quad c_6 = 0$$

and so on. Thus, two solutions are

$$y_1 = 1 - \frac{1}{2}x^2 + \frac{1}{720}x^6 + \cdots \quad \text{and} \quad y_2 = x - \frac{1}{6}x^3 - \frac{1}{60}x^5 + \cdots.$$

15. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned}
 y'' + xy' + 2y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} nc_n x^n}_{k=n} + 2 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\
 &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k + \sum_{k=1}^{\infty} kc_k x^k + 2 \sum_{k=0}^{\infty} c_k x^k \\
 &= 2c_2 + 2c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + (k+2)c_k]x^k = 0.
 \end{aligned}$$

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Thus

$$\begin{aligned} 2c_2 + 2c_0 &= 0 \\ (k+2)(k+1)c_{k+2} + (k+2)c_k &= 0 \end{aligned}$$

and

$$\begin{aligned} c_2 &= -c_0 \\ c_{k+2} &= -\frac{1}{k+1}c_k, \quad k = 1, 2, 3, \dots \end{aligned}$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$\begin{aligned} c_2 &= -1 \\ c_3 &= c_5 = c_7 = \dots = 0 \\ c_4 &= \frac{1}{3} \\ c_6 &= -\frac{1}{15} \end{aligned}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$\begin{aligned} c_2 &= c_4 = c_6 = \dots = 0 \\ c_3 &= -\frac{1}{2} \\ c_5 &= \frac{1}{8} \\ c_7 &= -\frac{1}{48} \end{aligned}$$

and so on. Thus, the general solution is

$$y = C_0 \left(1 - x^2 + \frac{1}{3}x^4 - \frac{1}{15}x^6 + \dots \right) + C_1 \left(x - \frac{1}{2}x^3 + \frac{1}{8}x^5 - \frac{1}{48}x^7 + \dots \right)$$

and

$$y' = C_0 \left(-2x + \frac{4}{3}x^3 - \frac{2}{5}x^5 + \dots \right) + C_1 \left(1 - \frac{3}{2}x^2 + \frac{5}{8}x^4 - \frac{7}{48}x^6 + \dots \right).$$

Setting $y(0) = 3$ and $y'(0) = -2$ we find $c_0 = 3$ and $c_1 = -2$. Therefore, the solution of the initial-value problem is

$$y = 3 - 2x - 3x^2 + x^3 + x^4 - \frac{1}{4}x^5 - \frac{1}{5}x^6 + \frac{1}{24}x^7 + \dots$$

16. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} (x+2)y'' + 3y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-1}}_{k=n-1} + 2 \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + 3 \underbrace{\sum_{n=0}^{\infty} c_n x^n}_{k=n} \\ &= \sum_{k=1}^{\infty} (k+1)kc_{k+1}x^k + 2 \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}x^k + 3 \sum_{k=0}^{\infty} c_k x^k \\ &= 4c_2 + 3c_0 + \sum_{k=1}^{\infty} [(k+1)kc_{k+1} + 2(k+2)(k+1)c_{k+2} + 3c_k]x^k = 0. \end{aligned}$$

Thus

$$\begin{aligned} 4c_2 + 3c_0 &= 0 \\ (k+1)kc_{k+1} + 2(k+2)(k+1)c_{k+2} + 3c_k &= 0 \end{aligned}$$

and

$$c_2 = -\frac{3}{4}c_0$$

$$c_{k+2} = -\frac{k}{2(k+2)}c_{k+1} - \frac{3}{2(k+2)(k+1)}c_k, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = -\frac{3}{4}$$

$$c_3 = \frac{1}{8}$$

$$c_4 = \frac{1}{16}$$

$$c_5 = -\frac{9}{320}$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we obtain

$$c_2 = 0$$

$$c_3 = -\frac{1}{4}$$

$$c_4 = \frac{1}{16}$$

$$c_5 = 0$$

and so on. Thus, the general solution is

$$y = C_0 \left(1 - \frac{3}{4}x^2 + \frac{1}{8}x^3 + \frac{1}{16}x^4 - \frac{9}{320}x^5 + \dots \right) + C_1 \left(x - \frac{1}{4}x^3 + \frac{1}{16}x^4 + \dots \right)$$

and

$$y' = C_0 \left(-\frac{3}{2}x + \frac{3}{8}x^2 + \frac{1}{4}x^3 - \frac{9}{64}x^4 + \dots \right) + C_1 \left(1 - \frac{3}{4}x^2 + \frac{1}{4}x^3 + \dots \right).$$

Setting $y(0) = 0$ and $y'(0) = 1$ we find $c_0 = 0$ and $c_1 = 1$. Therefore, the solution of the initial-value problem is

$$y = x - \frac{1}{4}x^3 + \frac{1}{16}x^4 + \dots$$

17. The singular point of $(1 - 2 \sin x)y'' + xy = 0$ closest to $x = 0$ is $\pi/6$. Hence a lower bound is $\pi/6$.
18. While we can find two solutions of the form

$$y_1 = c_0[1 + \dots] \quad \text{and} \quad y_2 = c_1[x + \dots],$$

the initial conditions at $x = 1$ give solutions for c_0 and c_1 in terms of infinite series. Letting $t = x - 1$ the initial-value problem becomes

$$\frac{d^2y}{dt^2} + (t+1)\frac{dy}{dt} + y = 0, \quad y(0) = -6, \quad y'(0) = 3.$$

Substituting $y = \sum_{n=0}^{\infty} c_n t^n$ into the differential equation, we have

$$\begin{aligned} \frac{d^2y}{dt^2} + (t+1)\frac{dy}{dt} + y &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n t^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n t^n}_{k=n} + \underbrace{\sum_{n=1}^{\infty} n c_n t^{n-1}}_{k=n-1} + \underbrace{\sum_{n=0}^{\infty} c_n t^n}_{k=n} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2}t^k + \sum_{k=1}^{\infty} k c_k t^k + \sum_{k=0}^{\infty} (k+1)c_{k+1}t^k + \sum_{k=0}^{\infty} c_k t^k \\ &= 2c_2 + c_1 + c_0 + \sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} + (k+1)c_{k+1} + (k+1)c_k]t^k = 0. \end{aligned}$$

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Thus

$$2c_2 + c_1 + c_0 = 0$$

$$(k+2)(k+1)c_{k+2} + (k+1)c_{k+1} + (k+1)c_k = 0$$

and

$$c_2 = -\frac{c_1 + c_0}{2}$$

$$c_{k+2} = -\frac{c_{k+1} + c_k}{k+2}, \quad k = 1, 2, 3, \dots$$

Choosing $c_0 = 1$ and $c_1 = 0$ we find

$$c_2 = -\frac{1}{2}, \quad c_3 = \frac{1}{6}, \quad c_4 = \frac{1}{12},$$

and so on. For $c_0 = 0$ and $c_1 = 1$ we find

$$c_2 = -\frac{1}{2}, \quad c_3 = -\frac{1}{6}, \quad c_4 = \frac{1}{6},$$

and so on. Thus, the general solution is

$$y = c_0 \left[1 - \frac{1}{2}t^2 + \frac{1}{6}t^3 + \frac{1}{12}t^4 + \dots \right] + c_1 \left[t - \frac{1}{2}t^2 - \frac{1}{6}t^3 + \frac{1}{6}t^4 + \dots \right].$$

The initial conditions then imply $c_0 = -6$ and $c_1 = 3$. Thus the solution of the initial-value problem is

$$y = -6 \left[1 - \frac{1}{2}(x-1)^2 + \frac{1}{6}(x-1)^3 + \frac{1}{12}(x-1)^4 + \dots \right]$$

$$+ 3 \left[(x-1) - \frac{1}{2}(x-1)^2 - \frac{1}{6}(x-1)^3 + \frac{1}{6}(x-1)^4 + \dots \right].$$

19. Writing the differential equation in the form

$$y'' + \left(\frac{1 - \cos x}{x} \right) y' + xy = 0,$$

and noting that

$$\frac{1 - \cos x}{x} = \frac{x}{2} - \frac{x^3}{24} + \frac{x^5}{720} - \dots$$

is analytic at $x = 0$, we conclude that $x = 0$ is an ordinary point of the differential equation.

20. Writing the differential equation in the form

$$y'' + \left(\frac{x}{e^x - 1 - x} \right) y = 0$$

and noting that

$$\frac{x}{e^x - 1 - x} = \frac{2}{x} - \frac{2}{3} + \frac{x}{18} + \frac{x^2}{270} - \dots$$

we see that $x = 0$ is a singular point of the differential equation. Since

$$x^2 \left(\frac{x}{e^x - 1 - x} \right) = 2x - \frac{2x^2}{3} + \frac{x^3}{18} + \frac{x^4}{270} - \dots,$$

we conclude that $x = 0$ is a regular singular point.

21. Substituting $y = \sum_{n=0}^{\infty} c_n x^n$ into the differential equation we have

$$\begin{aligned} y'' + x^2 y' + 2xy &= \underbrace{\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2}}_{k=n-2} + \underbrace{\sum_{n=1}^{\infty} n c_n x^{n+1}}_{k=n+1} + 2 \underbrace{\sum_{n=0}^{\infty} c_n x^{n+1}}_{k=n+1} \\ &= \sum_{k=0}^{\infty} (k+2)(k+1)c_{k+2} x^k + \sum_{k=2}^{\infty} (k-1)c_{k-1} x^k + 2 \sum_{k=1}^{\infty} c_{k-1} x^k \\ &= 2c_2 + (6c_3 + 2c_0)x + \sum_{k=2}^{\infty} [(k+2)(k+1)c_{k+2} + (k+1)c_{k-1}]x^k = 5 - 2x + 10x^3. \end{aligned}$$

Thus, equating coefficients of like powers of x gives

$$\begin{aligned} 2c_2 &= 5 \\ 6c_3 + 2c_0 &= -2 \\ 12c_4 + 3c_1 &= 0 \\ 20c_5 + 4c_2 &= 10 \\ (k+2)(k+1)c_{k+2} + (k+1)c_{k-1} &= 0, \quad k = 4, 5, 6, \dots, \end{aligned}$$

and

$$\begin{aligned} c_2 &= \frac{5}{2} \\ c_3 &= -\frac{1}{3}c_0 - \frac{1}{3} \\ c_4 &= -\frac{1}{4}c_1 \\ c_5 &= \frac{1}{2} - \frac{1}{5}c_2 = \frac{1}{2} - \frac{1}{5}\left(\frac{5}{2}\right) = 0 \\ c_{k+2} &= -\frac{1}{k+2}c_{k-1}. \end{aligned}$$

Using the recurrence relation, we find

$$\begin{aligned} c_6 &= -\frac{1}{6}c_3 = \frac{1}{3 \cdot 6}(c_0 + 1) = \frac{1}{3^2 \cdot 2!}c_0 + \frac{1}{3^2 \cdot 2!} \\ c_7 &= -\frac{1}{7}c_4 = \frac{1}{4 \cdot 7}c_1 \\ c_8 &= c_{11} = c_{14} = \dots = 0 \\ c_9 &= -\frac{1}{9}c_6 = -\frac{1}{3^3 \cdot 3!}c_0 - \frac{1}{3^3 \cdot 3!} \\ c_{10} &= -\frac{1}{10}c_7 = -\frac{1}{4 \cdot 7 \cdot 10}c_1 \\ c_{12} &= -\frac{1}{12}c_9 = \frac{1}{3^4 \cdot 4!}c_0 + \frac{1}{3^4 \cdot 4!} \\ c_{13} &= -\frac{1}{13}c_{10} = \frac{1}{4 \cdot 7 \cdot 10 \cdot 13}c_1 \end{aligned}$$

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and so on. Thus

$$\begin{aligned} y = c_0 & \left[1 - \frac{1}{3}x^3 + \frac{1}{3^2 \cdot 2!}x^6 - \frac{1}{3^3 \cdot 3!}x^9 + \frac{1}{3^4 \cdot 4!}x^{12} - \dots \right] \\ & + c_1 \left[x - \frac{1}{4}x^4 + \frac{1}{4 \cdot 7}x^7 - \frac{1}{4 \cdot 7 \cdot 10}x^{10} + \frac{1}{4 \cdot 7 \cdot 10 \cdot 13}x^{13} - \dots \right] \\ & + \left[\frac{5}{2}x^2 - \frac{1}{3}x^3 + \frac{1}{3^2 \cdot 2!}x^6 - \frac{1}{3^3 \cdot 3!}x^9 + \frac{1}{3^4 \cdot 4!}x^{12} - \dots \right]. \end{aligned}$$

22. (a) From $y = -\frac{1}{u} \frac{du}{dx}$ we obtain

$$\frac{dy}{dx} = -\frac{1}{u} \frac{d^2u}{dx^2} + \frac{1}{u^2} \left(\frac{du}{dx} \right)^2.$$

Then $dy/dx = x^2 + y^2$ becomes

$$-\frac{1}{u} \frac{d^2u}{dx^2} + \frac{1}{u^2} \left(\frac{du}{dx} \right)^2 = x^2 + \frac{1}{u^2} \left(\frac{du}{dx} \right)^2,$$

so $\frac{d^2u}{dx^2} + x^2u = 0$.

- (b) The differential equation $u'' + x^2u = 0$ has the form (18) in the text with

$$\begin{aligned} 1 - 2a = 0 & \implies a = \frac{1}{2} \\ 2c - 2 = 2 & \implies c = 2 \\ b^2c^2 = 1 & \implies b = \frac{1}{2} \\ a^2 - p^2c^2 = 0 & \implies p = \frac{1}{4}. \end{aligned}$$

Then, by (19) in the text,

$$u = x^{1/2} \left[c_1 J_{1/4} \left(\frac{1}{2}x^2 \right) + c_2 J_{-1/4} \left(\frac{1}{2}x^2 \right) \right].$$

- (c) We have

$$\begin{aligned} y &= -\frac{1}{u} \frac{du}{dx} = -\frac{1}{x^{1/2}w(t)} \frac{d}{dx} x^{1/2}w(t) = -\frac{1}{x^{1/2}w} \left[x^{1/2} \frac{dw}{dt} \frac{dt}{dx} + \frac{1}{2}x^{-1/2}w \right] \\ &= -\frac{1}{x^{1/2}w} \left[x^{3/2} \frac{dw}{dt} + \frac{1}{2x^{1/2}} w \right] = -\frac{1}{2xw} \left[2x^2 \frac{dw}{dt} + w \right] = -\frac{1}{2xw} \left[4t \frac{dw}{dt} + w \right]. \end{aligned}$$

Now

$$\begin{aligned} 4t \frac{dw}{dt} + w &= 4t \frac{d}{dt} [c_1 J_{1/4}(t) + c_2 J_{-1/4}(t)] + c_1 J_{1/4}(t) + c_2 J_{-1/4}(t) \\ &= 4t \left[c_1 \left(J_{-3/4}(t) - \frac{1}{4t} J_{1/4}(t) \right) + c_2 \left(-\frac{1}{4t} J_{-1/4}(t) - J_{3/4}(t) \right) \right] + c_1 J_{1/4}(t) + c_2 J_{-1/4}(t) \\ &= 4c_1 t J_{-3/4}(t) - 4c_2 t J_{3/4}(t) = 2c_1 x^2 J_{-3/4} \left(\frac{1}{2}x^2 \right) - 2c_2 x^2 J_{3/4} \left(\frac{1}{2}x^2 \right), \end{aligned}$$

so

$$y = -\frac{2c_1 x^2 J_{-3/4}(\frac{1}{2}x^2) - 2c_2 x^2 J_{3/4}(\frac{1}{2}x^2)}{2x[c_1 J_{1/4}(\frac{1}{2}x^2) + c_2 J_{-1/4}(\frac{1}{2}x^2)]} = x \frac{-c_1 J_{-3/4}(\frac{1}{2}x^2) + c_2 J_{3/4}(\frac{1}{2}x^2)}{c_1 J_{1/4}(\frac{1}{2}x^2) + c_2 J_{-1/4}(\frac{1}{2}x^2)}.$$

Letting $c = c_1/c_2$ we have

$$y = x \frac{J_{3/4}(\frac{1}{2}x^2) - cJ_{-3/4}(\frac{1}{2}x^2)}{cJ_{1/4}(\frac{1}{2}x^2) + J_{-1/4}(\frac{1}{2}x^2)}.$$

- 23. (a)** Equations (10) and (24) of Section 5.3 in the text imply

$$Y_{1/2}(x) = \frac{\cos \frac{\pi}{2} J_{1/2}(x) - J_{-1/2}(x)}{\sin \frac{\pi}{2}} = -J_{-1/2}(x) = -\sqrt{\frac{2}{\pi x}} \cos x.$$

- (b)** From (15) of Section 5.3 in the text

$$I_{1/2}(x) = i^{-1/2} J_{1/2}(ix) \quad \text{and} \quad I_{-1/2}(x) = i^{1/2} J_{-1/2}(ix)$$

so

$$I_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sum_{n=0}^{\infty} \frac{1}{(2n+1)!} x^{2n+1} = \sqrt{\frac{2}{\pi x}} \sinh x$$

and

$$I_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \sum_{n=0}^{\infty} \frac{1}{(2n)!} x^{2n} = \sqrt{\frac{2}{\pi x}} \cosh x.$$

- (c)** Equation (16) of Section 5.3 in the text and part (b) imply

$$\begin{aligned} K_{1/2}(x) &= \frac{\pi}{2} \frac{I_{-1/2}(x) - I_{1/2}(x)}{\sin \frac{\pi}{2}} = \frac{\pi}{2} \left[\sqrt{\frac{2}{\pi x}} \cosh x - \sqrt{\frac{2}{\pi x}} \sinh x \right] \\ &= \sqrt{\frac{\pi}{2x}} \left[\frac{e^x + e^{-x}}{2} - \frac{e^x - e^{-x}}{2} \right] = \sqrt{\frac{\pi}{2x}} e^{-x}. \end{aligned}$$

- 24. (a)** Using formula (5) of Section 3.2 in the text, we find that a second solution of $(1-x^2)y'' - 2xy' = 0$ is

$$\begin{aligned} y_2(x) &= 1 \cdot \int \frac{e^{\int \frac{2x}{1-x^2} dx}}{1^2} dx = \int e^{-\ln(1-x^2)} dx \\ &= \int \frac{dx}{1-x^2} = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right), \end{aligned}$$

where partial fractions was used to obtain the last integral.

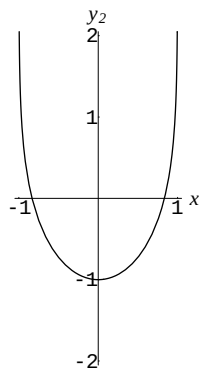
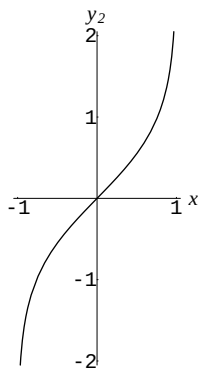
- (b)** Using formula (5) of Section 3.2 in the text, we find that a second solution of $(1-x^2)y'' - 2xy' + 2y = 0$ is

$$\begin{aligned} y_2(x) &= x \cdot \int \frac{e^{\int \frac{2x}{1-x^2} dx}}{x^2} dx = x \int \frac{e^{-\ln(1-x^2)}}{x^2} dx \\ &= x \int \frac{dx}{x^2(1-x^2)} dx = x \left[\frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) - \frac{1}{x} \right] = \frac{x}{2} \ln \left(\frac{1+x}{1-x} \right) - 1, \end{aligned}$$

where partial fractions was used to obtain the last integral.

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(c)



$$y_2(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

$$y_2 = \frac{x}{2} \ln \left(\frac{1+x}{1-x} \right) - 1$$

25. (a) By the binomial theorem we have

$$\begin{aligned} & [1 + (t^2 - 2xt)]^{-1/2} \\ &= 1 - \frac{1}{2}(t^2 - 2xt) + \frac{(-1/2)(-3/2)}{2!}(t^2 - 2xt)^2 + \frac{(-1/2)(-3/2)(-5/2)}{3!}(t^2 - 2xt)^3 + \dots \\ &= 1 - \frac{1}{2}(t^2 - 2xt) + \frac{3}{8}(t^2 - 2xt)^2 - \frac{5}{16}(t^2 - 2xt)^3 + \dots \\ &= 1 + xt + \frac{1}{2}(3x^2 - 1)t^2 + \frac{1}{2}(5x^3 - 3x)t^3 + \dots = \sum_{n=0}^{\infty} P_n(x)t^n. \end{aligned}$$

(b) Letting $x = 1$ in $(1 - 2xt + t^2)^{-1/2}$, we have

$$(1 - 2t + t^2)^{-1/2} = (1 - t)^{-1} = \frac{1}{1-t} = 1 + t + t^2 + t^3 + \dots \quad (|t| < 1) = \sum_{n=0}^{\infty} t^n.$$

From part (a) we have

$$\sum_{n=0}^{\infty} P_n(1)t^n = (1 - 2t + t^2)^{-1/2} = \sum_{n=0}^{\infty} t^n.$$

Equating the coefficients of corresponding terms in the two series, we see that $P_n(1) = 1$. Similarly, letting $x = -1$ we have

$$\begin{aligned} (1 + 2t + t^2)^{-1/2} &= (1 + t)^{-1} = \frac{1}{1+t} = 1 - t + t^2 - 3t^3 + \dots \quad (|t| < 1) \\ &= \sum_{n=0}^{\infty} (-1)^n t^n = \sum_{n=0}^{\infty} P_n(-1)t^n, \end{aligned}$$

so that $P_n(-1) = (-1)^n$.